

ALIBABA CLOUD INFRASTRUCTURE GUIDE FOR AWS PROFESSIONALS

 Alibaba Cloud

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AUTHORS

PRODUCER

Alex Chen

Senior Director of Infrastructural Cloud Services of Alibaba Cloud

CORE AUTHORS

Chen Yunlong, Jason Xu, Gu Haidong, Wang Shufan,
Mo Bin, Lin Yang, Ada Sang, Roger Wang, Liu Enze, Zhou Xiang,
Wang Zhongyu, Zhang Nan, Yang Wenting, Zhan Lifei, Hong Xiaolong,
Liu Rao, Li Chao

EXECUTIVE PRODUCERS

John Jiang

General Manager of Infrastructural Cloud Services of Alibaba Cloud

Selina Yuan

General Manager of Alibaba Cloud Intelligence International Business Unit

EDITING AND DESIGN

Sue Zhou and Brooks Lv

TARGET AUDIENCE

This guide is intended to provide Amazon Web services' (AWS) professionals with methodologies and best practices to understand Alibaba Cloud infrastructure, regardless of whether you are planning a multi-cloud solution with AWS and Alibaba Cloud or migrating to Alibaba Cloud.



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1. WHY MAKE THE MOVE?

Here are some key considerations that may drive customers to make the move from Amazon Web services (AWS) to Alibaba Cloud or to plan a multi-cloud solution comprising of both AWS and Alibaba Cloud.

1.1 COST EFFICIENCY

One of the major factors driving a migration off of AWS is cost efficiency. For the majority of the products on AWS, customers can find similar products on Alibaba Cloud with competitive prices, good performance, and high availability.

For example, as of November 2021, Alibaba Cloud offers their Elastic Compute Service (ECS) Virtual Machine instances with 99.975% availability for a single instance, which is the highest compared to other cloud providers. And at the same time, the list prices of many Alibaba Cloud ECS instances are cheaper than the AWS EC2 instances with similar specifications.

Furthermore, Alibaba Cloud offers flexible, easy-to-use, and cost effective resource packages to further save your cost, such as monthly subscriptions for many of the Alibaba Cloud products, Storage Capacity Unit (SCU) for storage, Data Transfer Plan for network traffic, and 95 percentile billing for Cloud Enterprise Network (CEN), etc. Find out more about [Alibaba Cloud pricing here](#).

1.2 GO CHINA & GO ASIA

Alibaba Cloud was born and raised in China and understands both the complexities and opportunities of China's market, ensuring that our customers are offered the outstanding "Go China" solutions. Please read more about [China Gateway here](#).

As a native Asian company and Asia's number one Infrastructure as a Service (IaaS) service provider recognized by Gartner¹, Alibaba Cloud would instantly accelerate your success in Asia. You can find out more about [Asia Accelerator here](#).





1.3 ECOSYSTEM

Alibaba Cloud has more than 10,000 global partners and connects with more than 10 business units within the Alibaba Group to offer a better service to customers, covering more than 200 industry solutions, including finance, retail, e-commerce, blockchain, media, and so on.

1.4 MULTI-CLOUD / HYBRID CLOUD

Alibaba Cloud is the pilot of hybrid clouds and has supported many customer successes across many industries as a result. Inherited from the Alibaba Cloud Public Cloud, the Alibaba Cloud Hybrid Cloud is equipped with over 50 full-stack product portfolios and solutions, enabling you to practice Hybrid Cloud in various business scenarios. Check out the Alibaba Cloud website to learn more about Alibaba Cloud's [Hybrid Cloud](#) solutions.

1.5 SECURITY AND COMPLIANCE

The security process of Alibaba Cloud has been recognized by authorities both inside and outside Mainland China. By leveraging its years of expertise in defending against Internet security threats, Alibaba Cloud provides security protection for the cloud platform and integrates multiple compliance standards into the internal control and product design of the cloud platform. Please read more about [Alibaba Cloud's security & privacy compliance here](#).

User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal states that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belong only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have the responsibilities and obligation to help users ensure the confidentiality, integrity, and availability of their data. More on [Alibaba Cloud's data security can be found here](#).

1.6 INSTANT SUPPORT

Alibaba Cloud offers highly flexible support services tailored to meet your exact needs. To better support enterprise customers, a dedicated technical account manager (TAM) would be assigned and a dedicated instant messaging DingTalk group would be created. Please read more about [Alibaba Cloud's after-sales support here](#).





2. SERVICE COMPARISON BETWEEN AWS AND ALIBABA CLOUD

As of November 2021, Alibaba Cloud has a total of 80 Zones strategically located across 25 regions in the world, providing reliable computing support for billions of customers. Alibaba Cloud has the most data centers (71 Zones located across 20 regions) in the Asia Pacific area than any other public cloud provider. Visit the Alibaba Cloud website to find out more about our [Data Centers](#) around the world.

Area	Country	City	Region Id	Number of Zones
Asia Pacific (19)	China	Beijing	cn-beijing	12
	China	Chengdu	cn-chengdu	2
	China	Guangzhou	cn-guangzhou	2
	China	Hangzhou	cn-hangzhou	8
	China	Heyuan	cn-heyuan	2
	China	Huhhot	cn-huhehaote	2
	China	Nanjing	cn-nanjing	1
	China	Qingdao	cn-qingdao	2
	China	Shanghai	cn-shanghai	9
	China	Shenzhen	cn-shenzhen	6
	China	Ulanqab	cn-wulanchabu	3
	China	Zhangjiakou	cn-zhangjiakou	3





Area	Country	City	Region Id	Number of Zones
Asia Pacific (19)	China	Hong Kong	cn-hongkong	3
	Japan	Tokyo	ap-northeast-1	2
	India	Mumbai	ap-south-1	2
	Singapore	Singapore	ap-southeast-1	3
	Australia	Sydney	ap-southeast-2	2
	Malaysia	Kuala Lumpur	ap-southeast-3	2
	Indonesia	Jakarta	ap-southeast-5	3
	Philippines	Manila	ap-southeast-6	2
Europe & Americas (4)	Germany	Frankfurt	eu-central-1	2
	UK	London	eu-west-1	2
	US	Silicon Valley	us-west-1	2
	US	Virginia	us-east-1	2
Middle East (1)	UAE	Dubai	me-east-1	1

Alibaba Cloud offers comprehensive services. For most of the services found on AWS, customers can find similar services on Alibaba Cloud. Below is a rough services comparison between AWS and Alibaba Cloud.

Please note not every AWS or Alibaba Cloud service is listed, and not every matched service has exact feature-for-feature parity. For more detailed information, please read this online [document](#).

2.1 ELASTIC COMPUTING

2.1.1 CLOUD SERVER

AWS Product	Alibaba Cloud Product	Description
Elastic Compute Cloud (EC2)	Elastic Compute Service (ECS)	Elastic and secure virtual cloud servers to cater to all your cloud hosting needs.



AWS Product	Alibaba Cloud Product	Description
EC2 Bare Metal Instances	ECS Bare Metal Instance (EBM)	An elastic and horizontally scalable high-performance computing service providing the same computing performance as traditional physical servers including physical isolation.
Lightsail	Simple Application Server	A single server-based service for application deployment, security management, operations & maintenance (O&M) monitoring, and more.
EC2 GPU Instances	Elastic GPU Service (EGS)	Powerful parallel computing capabilities based on Graphics Processing Unit (GPU) technology.
EC2 Dedicated Instances	Dedicated Host (DDH)	A dedicated resource pool that offers flexible and compliant deployment options to grant enterprise users more control over their cloud infrastructure.
Amazon WorkSpaces	Elastic Desktop Service (EDS)	A convenient and secure cloud-based Desktop-as-a-Service (DaaS) solution.

2.1.2 ELASTIC ORCHESTRATION

AWS Product	Alibaba Cloud Product	Description
Auto Scaling	Auto Scaling	Automatically adjusts computing resources based on your business cycle.
CloudFormation	Resource Orchestration Service (ROS)	A fully managed service for cloud computing resource orchestration and automated deployment.
Systems Manager	Operation Orchestration Service (OOS)	Automated operations and maintenance (O&M) service.

2.1.3 HIGH-PERFORMANCE COMPUTING

AWS Product	Alibaba Cloud Product	Description
High-Performance Computing	Elastic High Performance Computing (E-HPC)	A High Performance Computing as a Service (HPCaaS) cloud platform providing an all-in-one high-performance public computing service.
AWS Batch	Batch Compute	Resource management and task scheduling for large-scale batch processing.



2.2 NETWORKING

2.2.1 CLOUD NETWORK

AWS Product	Alibaba Cloud Product	Description
Amazon VPC	Virtual Private Cloud (VPC)	A virtual private cloud (VPC) service that provides an isolated cloud network to operate resources in a secure environment.
Elastic Load Balancing	Server Load Balancer (SLB)	Server Load Balancer (SLB) distributes network traffic across groups of backend servers to improve the service capability and application availability. It provides functions as a reverse proxy at Layer 7 (ALB – Application Load Balancer) and load balancing services at Layer 4 (CLB – Classic Load Balancer).
Private Link	PrivateLink	Connect your VPCs to services in other VPCs through secure, reliable, and private connections. PrivateLink simplifies network architecture and reduces security risks because network traffic does not traverse the Internet.
NAT Gateway	NAT Gateway	A NAT (Network Address Translation) Gateway enables multiple instances within a VPC to communicate with the Internet. Custom SNAT (Source Network Address Translation) and DNAT (Destination Network Address Translation) rules can be created to help you use network resources in a flexible manner.

2.2.2 CROSS-REGION NETWORK

AWS Product	Alibaba Cloud Product	Description
Transit Gateway	Cloud Enterprise Network (CEN)	A global network for rapidly building a distributed business system and hybrid cloud to help users create a network with enterprise-level scalability, with the communication capabilities of a cloud network.
AWS Global Accelerator	Global Accelerator (GA)	Provides a network acceleration service for your Internet-facing application globally with guaranteed bandwidth and high reliability.



2.2.3 HYBRID CLOUD NETWORK

AWS Product	Alibaba Cloud Product	Description
Direct Connect	Express Connect	An easy-to-use network service that enables high-bandwidth, reliability, security, and private connections between different environments.
VPN Gateway	VPN Gateway	A Virtual Private Network (VPN) Gateway is an Internet-based service that establishes a connection between a VPC and your on-premise data center.

2.3 STORAGE

2.3.1 DATA TRANSPORT

AWS Product	Alibaba Cloud Product	Description
AWS Snowball Edge Storage Optimized	Data Transport	A secure solution to migrate TB-level or PB-level data to Alibaba Cloud.

2.3.2 ESSENTIAL STORAGE SERVICE

AWS Product	Alibaba Cloud Product	Description
Amazon Elastic Block Store (EBS)	Elastic Block Storage (EBS)	Block-level data storage attached to ECS instances to achieve high performance, low latency, and high reliability.
Amazon Simple Storage Service (S3)	Object Storage Service (OSS)	An encrypted and secure cloud storage service that can store, process, and access vast amounts of data from anywhere in the world.
Amazon Elastic File System (EFS)	File Storage NAS	Simple, scalable, on-demand, and reliable network-attached storage (NAS) for use with ECS instances, HPC, and Container Service.
Amazon DynamoDB	Tablestore	A fully managed NoSQL cloud database service that enables the storage of a large amounts of structured and semi-structured data.

2.3.3 HYBRID CLOUD STORAGE

AWS Product	Alibaba Cloud Product	Description
AWS Storage Gateway	Cloud Storage Gateway (CSG)	Cloud Storage Gateway uses OSS for cloud-based storage at the back end and supports standard file and block storage protocols in the industry.

2.3.4 STORAGE DATA SERVICE

AWS Product	Alibaba Cloud Product	Description
AWS Cloud Watch, Cloud Watch Logs, Kinesis, Kinesis Firehose, Kinesis Analytics, Elastic Search	Log Service (SLS)	All-in-one service for log-type data.
AWS Backup	Hybrid Backup Recovery (HBR)	An enterprise-level data backup and migration service that is secure, efficient, cost-effective, and scalable. HBR provides full protection for data stored in Alibaba Cloud and on-premises data centers.

2.4 DATABASE

2.4.1 DATA WAREHOUSE

AWS Product	Alibaba Cloud Product	Description
Amazon Redshift	AnalyticDB for MySQL	A real-time data warehousing service that can process PB-level data with high concurrency and low latency.
Amazon Redshift	AnalyticDB for PostgreSQL	An online MPP (Massively Parallel Processing) data warehousing service based on the open-source Greenplum Database. It provides an online expansion and performance monitoring service to free your team from complicated MPP cluster operations and management (O&M).



AWS Product	Alibaba Cloud Product	Description
Amazon Athena	Data Lake Analytics (DLA)	Alibaba Cloud's DLA is an interactive analytics service that utilizes serverless architecture. As a ready-to-use service, DLA does not require any prior setup of infrastructure or upfront management costs. You do not need to maintain instances in DLA, and service is billed based on actual use and needs. DLA uses SQL (Structured Query Language) interfaces to interact with user service clients, which means it complies with standard SQL syntax and provides a variety of similar functions. DLA allows you to retrieve and analyze data from multiple data sources or locations such as OSS and Tablestore for optimal data processing, analytics, and visualization to give better insights and ultimately guide better decision-making.

2.4.2 NOSQL DATABASE

AWS Product	Alibaba Cloud Product	Description
Amazon ElastiCache for Redis	ApsaraDB for Redis	A fully managed, in-memory database service that is compatible with Redis protocols. It can process data in key-value pairs with low response latency. It is a popular choice for scenarios in which high concurrency and low response latency are necessary.
Amazon DocumentDB	ApsaraDB for MongoDB	A MongoDB-compatible database service that is developed based on the distributed system and high-reliability storage engine of Apsara. It uses a multi-node architecture to ensure high availability and supports elastic scaling, disaster recovery, backup and restoration, and performance optimization.
Amazon Timestream	Time Series Database (TSDB)	A highly reliable and cost-effective time series database that provides efficient data reading and writing capabilities, high-compression-ratio storage, and time series data interpolation and aggregation. It has wide industrial applications including Internet of Things (IoT) monitoring systems, enterprise-level energy management systems (EMS), production safety monitoring, and electric power detection systems.
Amazon Neptune	Graph Database (coming soon)	Fast and reliable Graph Database with TinkerPop Gremlin interface and ACID (Atomicity, Consistency, Isolation, Durability) property.

AWS Product	Alibaba Cloud Product	Description
Apache HBase on Amazon EMR	ApsaraDB for HBase	A cost-effective cloud-based intelligent NoSQL service that is of high scalability and compatible with the open-source HBase. This service provides core benefits such as low storage costs, high throughput and scalability, and intelligent data processing. It also ensures high availability by deploying data centers across regions.
Amazon Keyspaces (for Apache Cassandra)	ApsaraDB for Cassandra	A database engine fully compatible with Apache Cassandra with enterprise-level Service Level Agreement (SLA) assurance. It adopts a distributed, masterless architecture that is built for scaling and offers continuous uptime, always-online support. The database engine supports both long and wide tables with tens of thousands of columns and provides flexible schema capabilities.
Amazon Quantum Ledger Database (QLDB)	LedgerDB	LedgerDB provides blockchain database services. It takes advantage of blockchains and centralized systems. It also provides powerful data audit capabilities and is suitable for a wide variety of scenarios, such as data notarization and data provenance.

2.4.3 RELATIONAL DATABASE

AWS Product	Alibaba Cloud Product	Description
AWS Aurora	PolarDB	A new generation of "Super MySQL, PostgreSQL, or compatible Oracle" supporting up to 100TB storage.
Amazon RDS for MySQL	ApsaraDB RDS for MySQL	ApsaraDB RDS (Relational Database Service) for MySQL is an availability and performance enhanced MySQL service supporting versions 5.5, 5.6, 5.7, and 8.0.
Amazon RDS for PostgreSQL	ApsaraDB RDS for PostgreSQL	An on-demand PostgreSQL service with automated monitoring, backup, and disaster recovery capabilities.
Amazon RDS for SQL Server	ApsaraDB RDS for SQL Server	An on-demand database hosting service for SQL Server with automated monitoring, backup, and disaster recovery capabilities.
Amazon RDS for MariaDB	ApsaraDB RDS for MariaDB TX	ApsaraDB for MariaDB TX is compatible with Oracle and designed with multiple enterprise database features. It uses multiple storage engines, including MySQL InnoDB, to meet different user requirements.



2.4.4 UTILITY & TOOLS

AWS Product	Alibaba Cloud Product	Description
AWS Database Migration Service	Data Transmission Service (DTS)	Alibaba Cloud Data Transmission Service (DTS) helps you migrate data between data stores, such as relational databases, NoSQL databases, and data warehouses. You can use DTS to migrate your data to Alibaba Cloud or between combinations of cloud and on-premises data systems. DTS supports several data replication modes, including data migration, data integration, data synchronization, and change tracking. You can choose a combination of data replication modes that best suit your use cases.
AWS Application Migration Service (AWS MGN)	Advanced Database & Application Migration (ADAM)	Advanced Database and Application Migration (ADAM) service helps migrate database applications (especially for Oracle) to the cloud. ADAM has been a feature in the DTS (Data Transmission Service).

There are more database centric cloud solution hands-on for reference: <https://github.com/alibabacloud-howto/database>.

2.5 SECURITY

2.5.1 BUSINESS SECURITY

AWS Product	Alibaba Cloud Product	Description
AWS Rekognition	Content Moderation	Image and video moderation service that accurately detects inappropriate content.

2.5.2 RELATIONAL DATABASE

AWS Product	Alibaba Cloud Product	Description
AWS Shield - Advanced	Anti-DDoS	Comprehensive DDoS (distributed denial-of-service) protection for enterprises to intelligently defend sophisticated DDoS attacks, reduce business loss risks, and mitigate potential security threats.
AWS Shield - Standard	Anti-DDoS Basic	A cloud-based security service that protects your data and application from DDoS attacks.



AWS Product	Alibaba Cloud Product	Description
AWS WAF	Web Application Firewall (WAF)	An easy-to-use service that provides real-time monitoring of servers to ensure high availability.
AWS Certificate Manager	Alibaba Cloud SSL Certificates Service	Alibaba Cloud SSL (Secure Sockets Layer) Certificates Service allows customers to directly apply, purchase, and manage SSL certificates.
Amazon Guard Duty, AWS Security Hub	Security Center	A unified security management system that identifies, analyzes, and notifies you of security threats in real-time.
AWS Config	Cloud Config	A configuration audit service that tracks and audits configurations of your Alibaba Cloud resources, helping you achieve the compliance of resource configurations.
AWS Network Firewall	Cloud Firewall	Cloud Firewall is one of the first SaaS (Software as a Service) firewalls deployed on the public cloud. It is your first choice for network security that safeguards your businesses in the cloud.
AWS Systems Manager	Bastion Host	A unified, efficient, and secure platform that provides cloud-based O&M, access control, and operation audit.
AWS CloudTrail	ActionTrail	A service that monitors and records the actions of your Alibaba Cloud account, including the access to and use of Alibaba Cloud services using the Alibaba Cloud Management console, called application programming interface (API) operations, or software development kits (SDKs).

2.5.3 DATA SECURITY

AWS Product	Alibaba Cloud Product	Description
Amazon Macie	Sensitive Data Discovery and Protection (SDDP)	An all-in-one data security solution that provides various features, such as sensitive data detection, classification, grading, and de-identification, to help you meet compliance requirements specified in General Data Protection Regulation (GDPR) and personal information protection.
AWS Key Management Service (KMS)	Key Management Service (KMS)	Easily protect, manage, use, and audit your cryptographic keys on Alibaba Cloud.
AWS CloudHSM	Data Encryption Service	Industry-standard hardware security modules (HSMs) deployed on Alibaba Cloud.



2.5.4 IDENTITY MANAGEMENT

AWS Product	Alibaba Cloud Product	Description
AWS Identity and Access Management (IAM)	Resource Access Management (RAM)	Secure your cloud resources with Resource Access Management (RAM) to define fine-grained access permissions for users and groups.
AWS Identity Services	Identify-as-a-Service (IDaaS)	Make identity management a painless experience and eliminate Identity Silos.

2.6 CONTAINER/SERVERLESS/MIDDLEWARE

2.6.1 APPLICATIONS

AWS Product	Alibaba Cloud Product	Description
Amazon CloudWatch	Application Real-Time Monitoring Service (ARMS)	Build business monitoring capabilities with a real-time response based on front-end monitoring, application monitoring, and custom business monitoring capabilities.
AWS X-Ray	Tracing Analysis	Allows developers to quickly identify root causes and analyze performance bottlenecks for distributed applications.

2.6.2 CONTAINERS

AWS Product	Alibaba Cloud Product	Description
AWS Fargate	Elastic Container Instance (ECI), Serverless Container Service	An agile and secure serverless container instance service.
Amazon Elastic Kubernetes Service (EKS)	Container Service for Kubernetes (ACK)	A Kubernetes-based service that ensures high efficiency for enterprises by running containerized applications on the cloud.
Amazon Elastic Container Registry (ECR)	Container Registry (ACR)	A fully managed, cloud-native artifacts management platform. ACR provides end-to-end security, intelligent building, efficient distribution of Docker container images, Helm Charts, and Oracle Cloud Infrastructure (OCI) compatible artifacts.



AWS Product	Alibaba Cloud Product	Description
AWS App Mesh	Alibaba Cloud Service Mesh (ASM)	A fully managed service mesh platform that is compatible with Istio. ASM provides consistent traffic management, secure communication with authentication, and service observability in heterogeneous environments.

2.6.3 MESSAGE QUEUES

AWS Product	Alibaba Cloud Product	Description
Amazon Managed Streaming for Apache Kafka (Amazon MSK)	Message Queue for Apache Kafka	A fully managed Apache Kafka service to help you quickly build data pipelines for your big data analytics.
Amazon MQ	Message Queue for RabbitMQ	A distributed, fully managed, and professional messaging service that features high throughput, low latency, and high scalability. This service supports Advanced Message Queuing Protocol (AMQP) and is fully compatible with open-source RabbitMQ.
Amazon Simple Notification Service (SNS)	Message Service	A message queuing and notification service that facilitates the smooth transfer of messages between applications.

2.6.4 MICRO SERVICES

AWS Product	Alibaba Cloud Product	Description
AWS Elastic Beanstalk	Enterprise Distributed Application Service (EDAS)	A PaaS (Platform as a Service) platform for a variety of application deployment options and microservices solutions to help monitor, diagnose, operate, and maintain your applications.
AWS AppConfig	Application Configuration Management (ACM)	Centralizes the management of application configurations and performs real-time configuration push in a distributed environment.





2.6.5 SERVERLESS

AWS Product	Alibaba Cloud Product	Description
AWS Lambda	Function Compute (FC)	A secure and stable, elastically scaled, O&M-free, pay-as-you-go, serverless computing platform.
AWS Step Functions	Serverless Workflow	Visualization, O&M-free orchestration, and Coordination of Stateful Application Scenarios.





3. MIGRATION PROCESS

Moving from one cloud provider to another could present complicated progress. Therefore, Alibaba Cloud has summarized its methodology into five steps based on past experience with thousands of customers, so that the procedure can be easily followed.

STEP 1: ASSESSMENT

It's important to always start with an assessment to understand the system architecture (e.g. server inventory), migration feasibility (e.g. downtime and data requirement), and the potential risk associated with the migration.

STEP 2: POC

With that information, pilot testing can be carried out in the way of POC (Proof of Concept) to ensure that the product functionality can meet the business requirement.

STEP 3: DESIGN

Once POC is done, it's on to the design stage. This involves planning the deployment scheme, mapping the products used in AWS to Alibaba Cloud, making possible changes to the application, and planning the resource size needed.

STEP 4: IMPLEMENTATION

After the solution has been designed, implementation takes place, part by part; firstly, the network needs to be built along with the operation environment, and the applications need to be prepared in Alibaba Cloud. Historical data can be moved first, whereas transaction data can be ported over during the cutover. More details on this can be found in the product sessions.

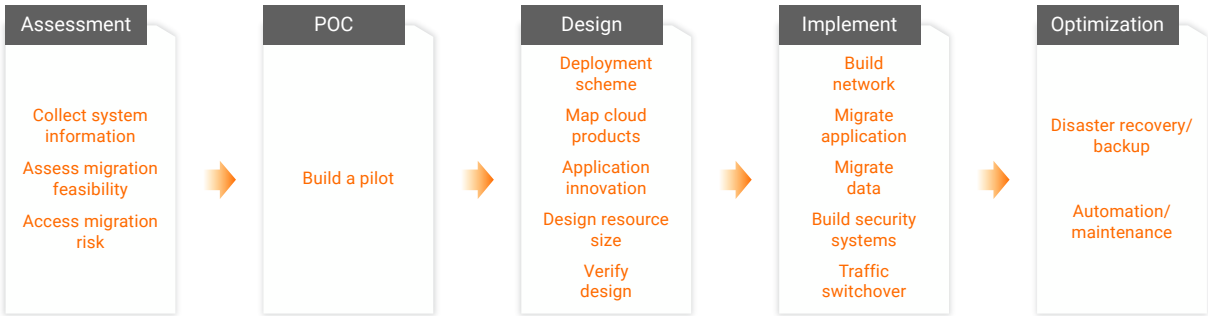




In addition to the implementation plan, a contingency plan should always be prepared in case of rollback.

STEP 5: OPTIMIZATION

It's always necessary to keep your setup optimized after implementation. This includes disaster recovery (DR) setup, backup, automation, and maintenance.

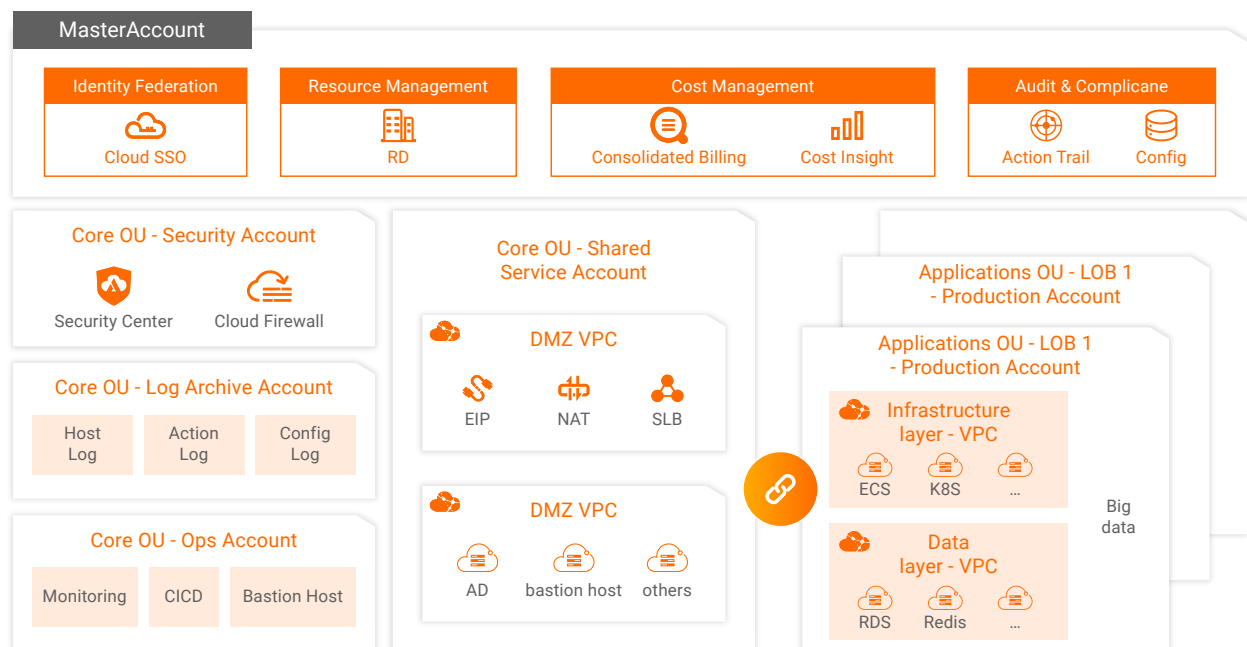




4. MIGRATION PRACTICES

4.1 CONSIDER YOUR ACCOUNT STRUCTURE AND IAM (IDENTITY AND ACCESS MANAGEMENT) - LANDING ZONE

In order to ensure the operational safety, compliance, manageability, and scalability of the cloud business, while accelerating the speed of cloud migration, it's recommended that customers design and build an Alibaba Cloud Landing Zone before migrating. When migrating, priority is given to the two neighborhoods of resource planning and identity authority. The typical architecture design is as follows:





4.1.1 MIGRATE MULTI-ACCOUNT STRUCTURE

Multi-account structure is the basis of Landing Zone. The following will explain how to set up Landing Zone multi-account structure in Alibaba Cloud.

1. Determining the master account

The first step is to confirm the master account of the enterprise. The enterprise management master account includes the Resource Directory (RD), which is used to manage multiple accounts. You can use this account as a financial custody account, to make unified payments and settlements for other accounts, and to view cloud bills in a unified manner. Unify the audit rules configured by this account and send them to each member account.

2. Creating Resource Directory

Alibaba Cloud Resource Directory can provide AWS Organizations of the same capacity, nouns corresponding relationship is as follows:

AWS Organizations	Alibaba Cloud Resource Directory
Organizational Units (OUs)	Folder
AWS Accounts	Member
AWS Resource Access Manager	Resource Sharing

You can use the master account to enable the Resource Directory service and create a resource folder in the Resource Directory based on the actual service conditions of your enterprise. You can create at least two resource directories consistent with AWS usage:

- » Core: Centrally manages public service accounts
- » Custom: Centrally manages service accounts

3. Creating a public service account

After the Resource Directory is set up, you need to create three shared service accounts. The process of creating accounts is almost the same as AWS, which can be completed in the resource catalog. It should be noted that the functions provided by these public service accounts are realized by different products in Alibaba Cloud.





a. Shared service account

Corporate shared services will be deployed in this account, such as network configuration, unified access to the public network configuration, etc.

For network-related cloud products, refer to the network section.

b. Log account

This aggregates the logs of all member accounts. Including cloud product operation behavior logs, audit logs, and business logs.

AWS	Alibaba Cloud
AWS Cloud Trail	Action Tail
AWS Config	Cloud Config

c. Security account:

Generally, the security role in the enterprise will use this account to configure related security products, such as common web application firewall (WAF), high-defense IP, etc.

For security-related cloud products, refer to the security section.

4.1.2 ACCESS CONTROL

Alibaba Cloud uses products and services such as access control (RAM or Resource Access Management) to achieve identity management and access control capabilities.

RAM is an advanced access control mechanism based on Attribute-Based Access Control (ABAC). When a customer request arrives, Alibaba Cloud will evaluate the currently accessed request characteristics, identity characteristics, and resource characteristics, and match them with the configured permissions of the identity to complete the authentication.

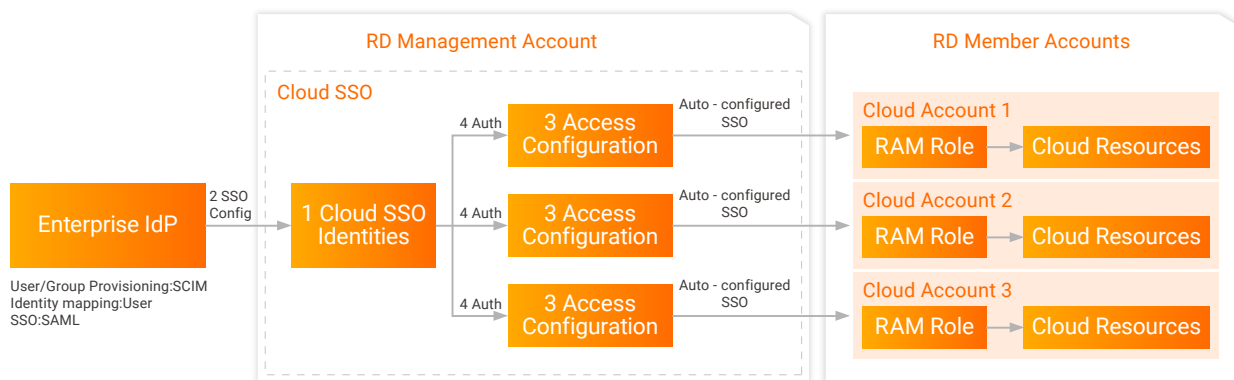
The identity and access control granularity, which is supported by each cloud product, can be found in cloud services that [support RAM](#) and cloud services that [support STS \(Security Token Service\)](#).



4.1.3 SINGLE SIGN-ON (SSO)

In the multi-account scenario, each account needs to manage its identity permissions and configure the single sign-on between each account and Alibaba Cloud. This management cost is very high.

We recommend using Alibaba Cloud's [CloudSSO](#), which can integrate the multi-account unified identity management and access control of the Alibaba Cloud Resource Directory (RD).

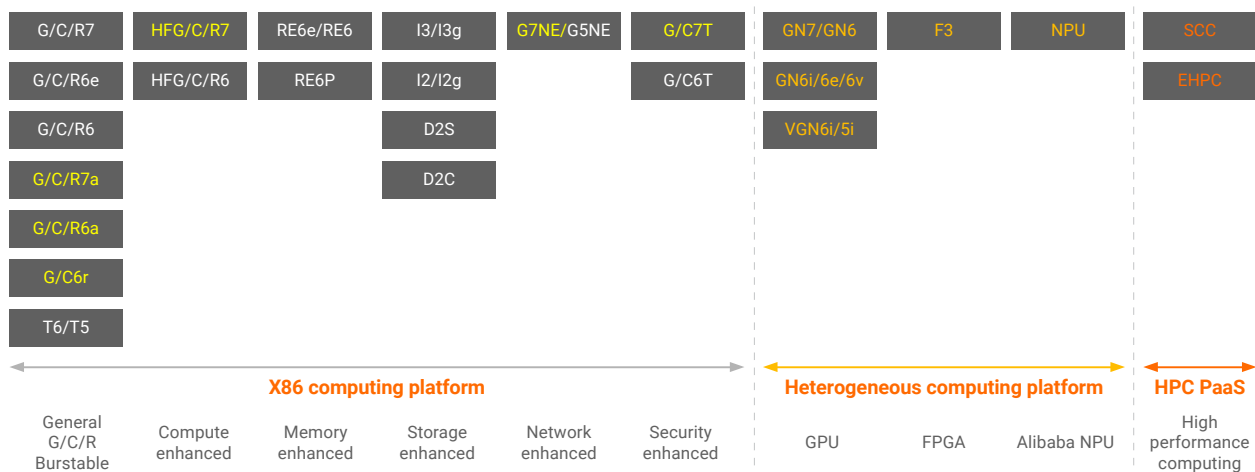


4.2 COMPUTE SERVICE

4.2.1 CHOOSE ECS INSTANCE TYPE ACCORDING TO AWS EC2

Before migrating AWS VM to Alibaba Cloud, you have to choose the corresponding instance type carefully. With appropriate instance types, it can improve the application performance while reducing the total cost of computing resources.

Below is a diagram showing the Alibaba Cloud instance families portfolio. You can find what types of workloads or scenarios are running on each instance family on the Alibaba Cloud [website](#).



Here is Alibaba Cloud's ECS instance type compared with AWS instance type side by side, which provides a brief guideline on how to select instance families.

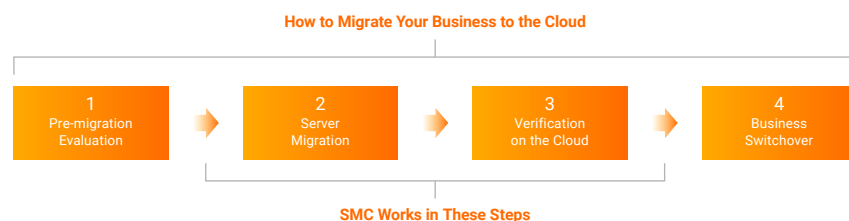
Instance Description	AWS	Alibaba Cloud
Burstable	T4g/T3a/T3/T2	T6/T5
General Purpose (Intel CPU)	M6i/M5/M4	G7/G6e/G6/G5 HFG7/HFG6
Compute Optimized (Intel CPU)	C5/C4	C7/C6e/C6/C5 HFC7/HFC6
Memory Optimized (Intel CPU)	R5/R4/X1e/X1	R7/R6e/R6/R5 HFR7/HFR6
Memory Enhanced	u-6tb1.metal	RE6/RE4
Security Enhanced	Enclave	Enclave (G/C/R7) vSGX (G/C/R7t, G/C/R6t)
Instance with Local SSD	I3	I3/I3g/I2ne/I2gne
Instance with Local HDD	D2/H1	D2S/D2C
GPU Instances	P3/P2/inf1/G4/G3	GN7/GN6/GN5 vGN6i/vGN5i
FPGA Instances	F1	F3/F1
AMD Instances	M5a/C5a/R5a	G7a/C7a/R7a/G6a/C6a/R6a
ARM Instances	M6g C6g/R6g	G6r (Recommend C6/R6)

4.2.2 USING SMC TO MIGRATE VM FROM AWS TO ALIBABA CLOUD

What is SMC?

Alibaba Cloud Server Migration Center (SMC) is a free tool for Cloud Migration, which launched in November 2017. SMC allows you to migrate one or more source servers to Alibaba Cloud. Source servers can be servers in data centers, virtual machines, servers on other cloud platforms, and other types of servers. SMC helps you to automate the migration of server application environments and simplifies the migration of system images to ECS instances. To facilitate the management of migration tasks, SMC provides service-oriented API operations that allow you to create and execute migration scripts and query the migration progress.

How SMC can help?



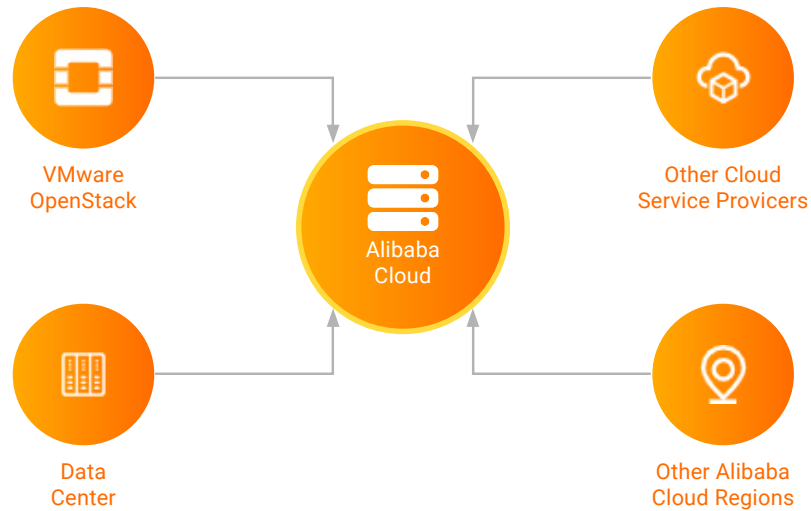
SMC works in server migration and can help to verify a VM after migration. Before SMC migrates to a VM, application pre-migration evaluation should be carried out. We suggest migrating your VM using the pay-as-you-go (PAYG) purchasing model before the business shift to Alibaba Cloud and, after that, you can change from

PAYG to other purchasing models, such as monthly or yearly subscriptions for those all-time running workloads.



NOTE

For database migration, DTS is recommended.

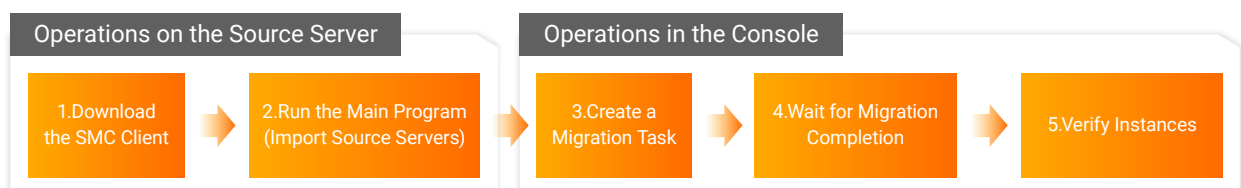


SMC allows you to migrate servers from your own data centers, your local virtual machines (VMware, Virtual Box, Xen, and KVM), and other cloud platforms such as AWS, Azure, Google Cloud, Huawei Cloud, Tencent Cloud, UCloud, Telecom Cloud, and QingCloud.

Key benefits of SMC



SMC migration steps



You have to make sure the following things are well prepared before you perform SMC migration:

4. RAM and AccessKey are prepared <https://www.alibabacloud.com/help/doc-detail/122993.htm?spm=a2c63.p38356.b99.8.6ebe6708RXyGDw>
5. Network connection (either public or private network) established between source VM and destination VM.
6. Destination network environments created including VPC, vSwitch, and security group.

There are five steps to migrate a VM to Alibaba Cloud, which are as follows:

- » Step 1: Download and install the SMC client on your source server.
 - » The client has 32-bit and 64-bit versions for Windows and Linux (please ensure the correct one is chosen).
 - » "go2aliyun_client" is the main program.
 - » "user_config.json" is the user configuration file.
- » Step 2: Run the main program in the source server to import the source server information to Alibaba Cloud SMC service.
- » Step 3: Create a migration task in the Alibaba Cloud SMC console and start the job.
- » Step 4: Migration will run automatically until completion; the time spent depends on the source server data capacity and outbound bandwidth.
- » Step 5: Verify the migration result.

Here are some useful links for SMC tools and, if you have any trouble using this tool, you can contact the Alibaba Cloud GTS (Global Translation Service) team for support:

- » SMC Best Practice: <https://bp.aliyun.com/detail/14?accounttraceid=f15d772f7442418c84e052ac7677wiot>
- » SMC Console: <https://smc.console.aliyun.com/home>
- » SMC Help: <https://www.alibabacloud.com/help/product/121538.html>



4.3 NETWORKING

4.3.1 SLB

1. CLB deployment

CLB (Classic Load Balancer) is a product of Alibaba Cloud's load balancing product line. It provides layer 4 and basic layer 7 load balancing forwarding capabilities, corresponding to AWS's CLB products. The creation and use of CLB is very simple. You can use the console to create and manage CLB instances or you can use APIs to operate. The help documents for console operations can be found below:

<https://www.alibabacloud.com/help/doc-detail/86456.htm?spm=a2c63.l28256.b99.141.98782cc8jtCZvR>

2. Differences with AWS CLB

AWS will soon close the CLB product. However, as an alternative, you can use CLB in Alibaba Cloud for layer 4 and/or layer 7 traffic load balancing.

3. ALB deployment

ALB is the latest product of Alibaba Cloud's load balancing product line. It is a product specifically oriented to layer 7 application load balancing. It provides many powerful functions based on layer 7 services, both in terms of internet and ALB computing power. With the ability to automatically scale up and down, it can greatly simplify the configuration, deployment and planning of customers, and focus on application deployment. You can use the ALB console or API to create and manage ALB instances. The console operation help documents are as follows:

<https://www.alibabacloud.com/help/doc-detail/210822.htm?spm=a2c63.p38356.b99.13.428b3808CoKt4u>

4. Differences with AWS ALB

- a. AWS ALB does not support configuring QUIC (Quick UDP Internet Connection) listening, whereas Alibaba can support the creation of QUIC-based listening. This protocol can help you greatly improve the client access speed, especially in the frequent switching of weak networks, as well as Wi-Fi and mobile networks, which can realize multiplexing without reconnection, thus improving the access efficiency of resources and ensuring the security of data transmission.



- b.** AWS ALB does not support fixed IP (Internet Protocol) mode. ALB can support fixed IP and then the resilience is limited but can still meet the Domain Name System (DNS) resolution to fixed IP scenarios.

4.3.2 VPC

5. VPC deployment

Alibaba Cloud VPC is the counterpart for AWS VPC. To avoid conflict, it is suggested that you create an Alibaba Cloud VPC that has the different CIDR (Classless Inter-Domain Routing) Block to your current AWS VPC. This will definitely be helpful to the resource migration because both cloud platforms don't support layer 2 connection on a physical link. The console operation help documents are as follows:

<https://www.alibabacloud.com/help/doc-detail/65430.htm?spm=a2c63.p38356.b99.18.67856d4c1z0HOB>

4.3.3 NAT GATEWAY

1. NAT Gateway deployment

NAT Gateway (NATGW) is one of the basic network element products of Alibaba Cloud. It provides the function of IP NAT for the public and internal network, which can meet the mainstream network function requirements of customers. You can use the following documents to create and manage NATGW instances:

<https://www.alibabacloud.com/help/doc-detail/121139.htm?spm=a2c63.l28256.b99.25.5cda655cpjqrc4>

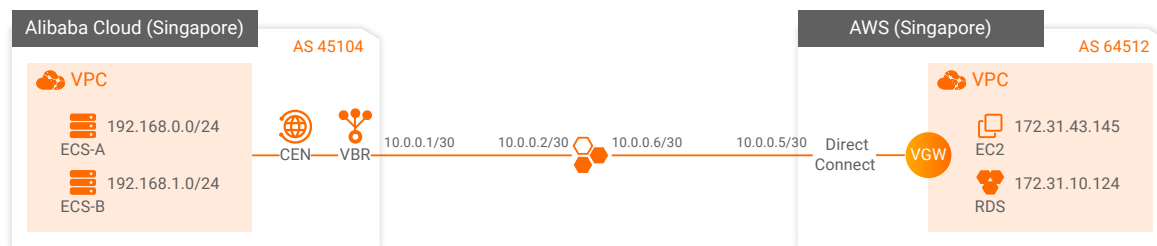
2. Differences with AWS NAT Gateway

- a.** AWS NATGW only supports SNAT, whereas Alibaba Cloud's NATGW supports both SNAT and DNAT. This allows Alibaba Cloud NATGW to accept VPC ingress traffic from the Internet.
- b.** AWS NATGW can only bind one Elastic IP address (EIP), whereas Alibaba Cloud's NATGW can bind multiple Elastic IP addresses. This allows you to create different SNAT rules that mapping different egress traffic by source CIDR block to different EIP.
- c.** You need to build SNAT or DNAT rules in Alibaba Cloud NATGW because it supports binding to multiple EIP as described above, while AWS don't have such configuration.



4.3.4 BUILD A PRIVATE CONNECTION BETWEEN ALIBABA CLOUD AND AWS

3. Express Connect



Whether it is Alibaba Cloud or AWS, there are many dedicated leased line partners who can provide dedicated interconnection between Alibaba Cloud and AWS (similar to China Unicom/China Mobile/China Telecom/Equinix, etc.). Customers can purchase multi-cloud connection products through third-party service providers for the dedicated line interconnection line interconnection between Alibaba Cloud and AWS. Physical links are generally divided into exclusive type and shared type. Lines of dedicated type require service providers to perform fiber construction through LOA (Letter of Authorization), which is usually in-building lines. The cycle is relatively long and two specifications of 1G/10G ports are provided. Lines of shared type are opened, pre-built, dedicated lines by Internet Service Provider (ISP) and can provide various specifications from 50Mbps to 1Gbps. The cycle is relatively short in comparison to the dedicated type.

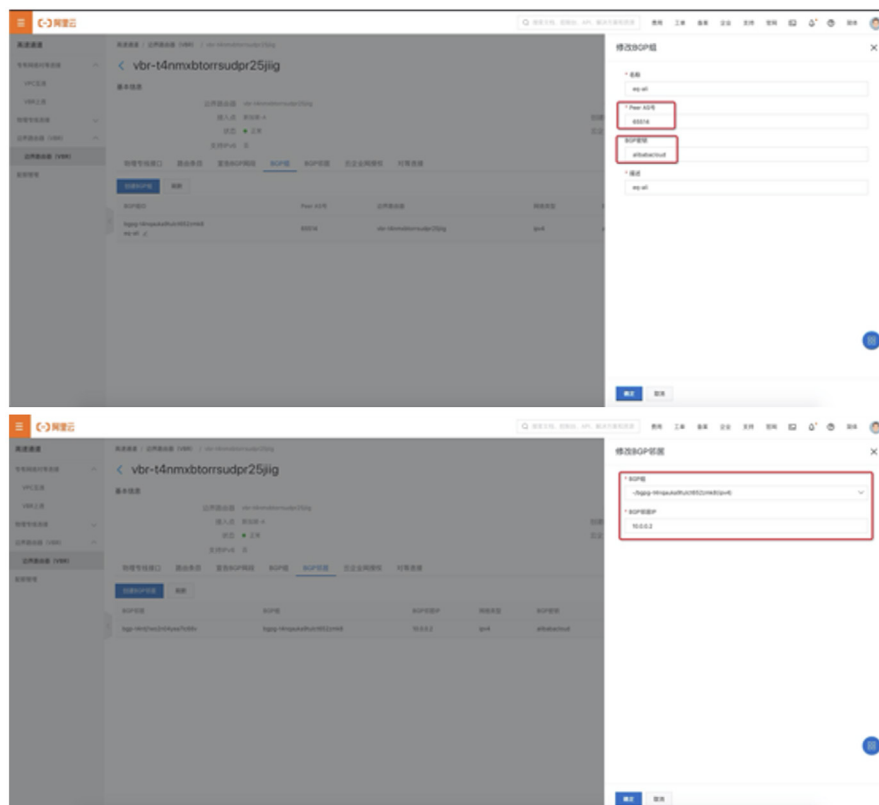
Configurations:

» Alibaba Cloud side

- a. After the creation of the Network Edge is complete, a Variable Bit Rate (VBR) instance will appear on the Aliyun console Express Connect - Virtual Border Router. Note that the Singapore region is selected in the upper left corner:



- b. Click on the VBR instance to create a Border Gateway Protocol (BGP) group and BGP neighbor:



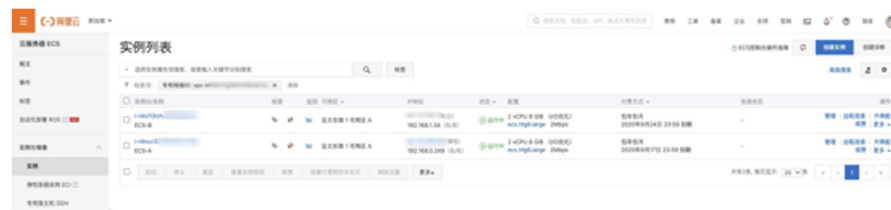
After the configuration is complete, you can see that the BGP negotiation status is Established:



- c. Create a VPC and a Cloud Enterprise Network (CEN) in the Singapore region, and load the VPC and VBR into the CEN:



- d. Create two ECSs in the VPC to simulate two different types of users:

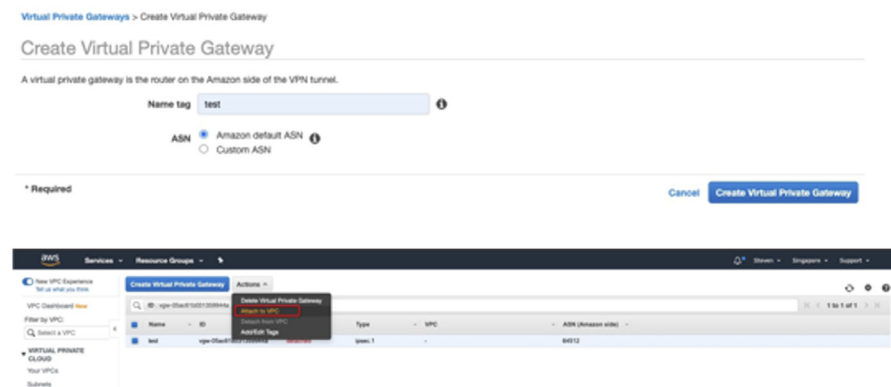


» AWS side

- a. You can see a connection in the AWS Direct Connect console:



- b. Create a Video Gateway (VGW) and attach it to the existing AWS VPC:



- c. Create a Virtual Interface (VIF) instance to enter the BGP configuration, but be sure to pay attention to select the VGW you just created:

Virtual interface type

Type

☒ **Private**
A private virtual interface should be used to access an Amazon VPC using private IP addresses.

☐ **Public**
A public virtual interface can access all AWS public services using public IP addresses.

☐ **Transit**
A transit virtual interface is a VLAN that transports traffic from a Direct Connect gateway to one or more transit gateways.

Private virtual interface settings

Virtual interface name
A name to help you identify the new virtual interface.
test
Name must contain no more than 100 characters. Valid characters are a-z, 0-9, and - (hyphen)

Connection
The physical connection on which the new virtual interface will be provisioned.
test2

Virtual interface owner
The account that will own the virtual interface.
☒ **My AWS account**
☐ Another AWS account

☐ **Transit**
A transit virtual interface is a VLAN that transports traffic from a Direct Connect gateway to one or more transit gateways.

Private virtual interface settings

Virtual interface name
A name to help you identify the new virtual interface.
test
Name must contain no more than 100 characters. Valid characters are a-z, 0-9, and - (hyphen)

Connection
The physical connection on which the new virtual interface will be provisioned.
test2

Virtual interface owner
The account that will own the virtual interface.
☒ **My AWS account**
☐ Another AWS account

Gateway type
Gateway type for this virtual interface.
☐ Direct Connect Gateway - recommended
Allows connections to multiple VPCs and regions.
☒ **Virtual Private Gateway**
Allows connections to a single VPC in the same region.

Virtual private gateway
A virtual private gateway attached to a VPC you wish to connect to.
test

VLAN
The Virtual Local Area Network number for the new virtual interface
348
Valid ranges are 1 - 4094

BGP ASN
The Border Gateway Protocol Autonomous System Number of your gateway for the new virtual interface.
65514
Valid ranges are 1 - 2147483647.

► Additional settings

Cancel Create virtual interface

Virtual Private Gateway
Allows connections to a single VPC in the same region

Virtual private gateway
A virtual private gateway attached to a VPC you wish to connect to.
test

VLAN
The Virtual Local Area Network number for the new virtual interface.
348
Valid ranges are 1 - 4094

BGP ASN
The Border Gateway Protocol Autonomous System Number of your gateway for the new virtual interface.
65514
Valid ranges are 1 - 2147483647.

Additional settings

Address family - optional
Determines whether the virtual interface is created with an IPv4 or IPv6 peering.
☒ IPv4
☐ IPv6

Your router peer ip - optional
The BGP peer IP configured on your endpoint.
10.0.0.6/30

Amazon router peer ip - optional
The BGP peer IP configured on the AWS endpoint.
10.0.0.5/30

BGP authentication key - optional
The password that will be used to authenticate the BGP session.
awscloud

Jumbo MTU (MTU size 9001) - optional
Allow MTU size of 9001 on virtual interface.
☐ Enabled

Tags
Specified tags to help identify a AWS Direct Connect resource.
No tags associated with the resource

Cancel Create virtual interface

After the configuration is complete, the status of Peering shows up:

General configuration

Virtual interface ID	Virtual interface name	Virtual private gateway	Virtual interface type	Region	Amazon VPC ID	Connection ID	Virtual interface type	Region
vxif-fgugl77p	vxif-fgugl77p	vgw-0a1b1c1d1e1f1g1h	private	ap-southeast-1	vxif-fgugl77p	vxif-fgugl77p	private	ap-southeast-1

Peering (1)

ID	Name	BGP ASN	BGP authentication key	Your router peer IP	Amazon router peer IP	Amazon VPC ID	Status	Peering status
vxif-fgugl77p	vxif-fgugl77p	65514	awscloud	10.0.0.6/30	10.0.0.5/30	vxif-fgugl77p	available	Peering status

The connection of AWS VIF to Network Edge is complete.

- d. Create EC2 in AWS VPC. Note that the subnet needs to be separated from the Alibaba Cloud side. This example selects 172.31.0.0/16

New EC2 Instance

Launch Instance

Instance ID: i-01234567890123456

Instance Type: t2.micro

Availability Zone: us-east-1a

Instance State: Pending

Status Checks: 2/2 checks passed

Public DNS (IPv4): ec2-172-31-0-100.us-east-1.compute.amazonaws.com

Private DNS (IPv4): 172.31.0.100

Key Name: my-key

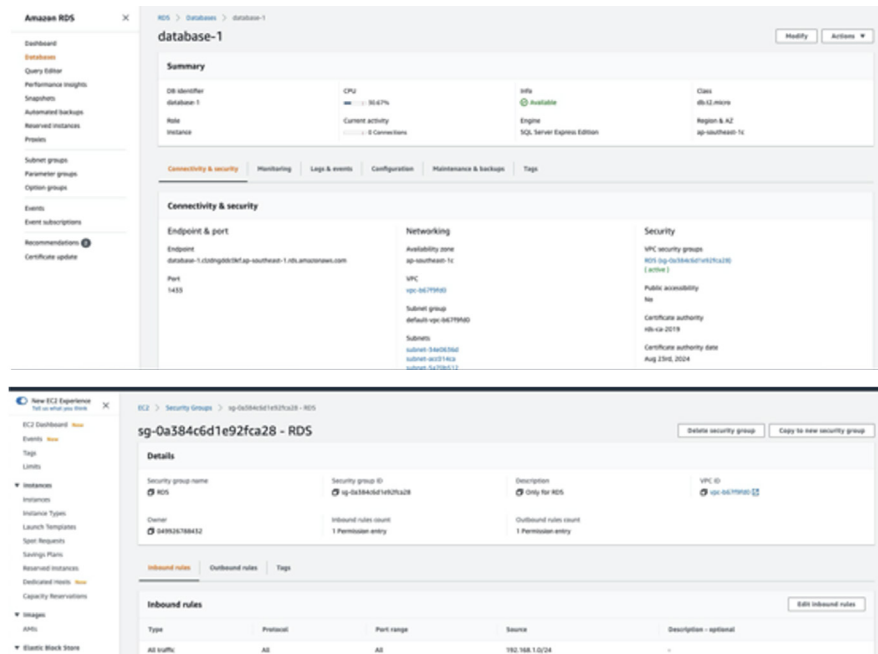
Security Group: sg-01234567

Subnet: subnet-01234567

Placement Group: default

Monitoring: Amazon CloudWatch

- e. Create RDS in AWS VPC. Due to business needs, ordinary users are not allowed to access AWS RDS on the 192.178.0.0/24 network segment. Pay attention to the security group settings here and only allow access to the 192.168.1.0/24 network segment:



2. VPN Gateway

VPN Gateway is an Internet-based service that securely and reliably connects enterprise data centers, office networks, and third-party public clouds, such as AWS or Azure, to VPCs of Alibaba Cloud through encrypted channels.

Configurations:

» Alibaba Cloud side:

<https://www.alibabacloud.com/help/doc-detail/110803.htm?spm=a2c63.p38356.b99.46.7fb925daz1YFzT>

» AWS side:

https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

3. Smart Access Gateway vCPE

SAG vCPE (Virtual Customer Premises Equipment) provides an image that can be deployed on data center servers, Edge Node Service (ENS) instances, Alibaba Cloud instances, Amazon Web Service (AWS) instances, and Microsoft Azure instances. After you deploy the SAG vCPE image on an instance, the instance serves as a vCPE device. The bandwidth of private networks for encrypted connections can reach 300 Mbit/s and higher (the

packet length in the performance test is 1,024 bytes). This allows you to connect private networks to Alibaba Cloud in a more flexible way.

Configurations:

<https://www.alibabacloud.com/help/doc-detail/179290.htm?spm=a2c63.p38356.b99.138.18004ef2lhvJ5d>

4. High Availability

a. Two physical leased lines

Two physical leased lines can work in an Active/Active or Active/Standby manner. Both Alibaba Cloud and AWS support the feature.

» Alibaba Cloud side:

<https://www.alibabacloud.com/help/doc-detail/143049.htm?spm=a2c63.l28256.b99.57.4aac6e07tl5AY6>

» AWS side:

<https://aws.amazon.com/directconnect/resiliency-recommendation/>

b. Physical leased line with VPN Gateway

You can configure the function of automatically switching to IPsec-VPN (Internet Protocol Security Virtual Private Network) when the physical leased line fails.

» Alibaba Cloud side:

<https://www.alibabacloud.com/help/zh/doc-detail/110823.htm?spm=a2c63.l28256.b99.91.fadb3c6c0rOhPz>

» AWS side:

<https://aws.amazon.com/cn/premiumsupport/knowledge-center/configure-vpn-backup-dx/>

c. Multiple VPN tunnels

You can configure multiple IPsec-VPN tunnels to realize the function of mutual backup.

» Alibaba Cloud side:

<https://www.alibabacloud.com/help/doc-detail/110821.htm?spm=a2c63.p38356.b99.89.5a991350qoCbAg>

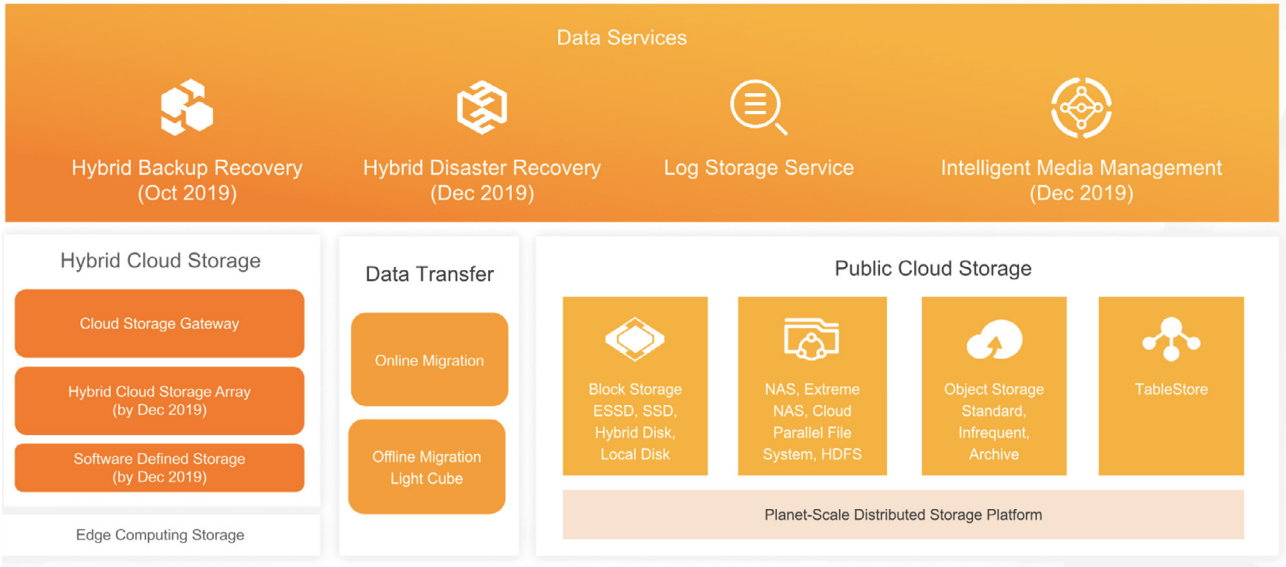
» AWS side:

<https://docs.aws.amazon.com/vpn/latest/s2svpn/vpn-redundant-connection.html>



4.4 STORAGE

For storage, Alibaba Cloud provides many products and services as below. The first step to consider is data migration for the foundational storage products.



4.4.1 DATA MIGRATION

When migrating from AWS, your data might come from the following sources:

Source	Destination
Object Storage in S3	OSS
Files in EFS	File Server NAS
Archive/Deep Archive data in Glacier	OSS Archive/Cold Archive

When you migrate your data from AWS to Alibaba Cloud or perform any cross-cloud data migration in general, you need to consider the following:

- » Data size and timeline. If the total data size is within a few TBs and the migration task is not time-sensitive, consider using online migration through the Internet. If the total data size is 100TB or above, consider using online migration through the express route. For example, the table below shows sample figures to see how long a migration job can take (for reference). In real scenarios, the time to migrate also depends on the actual bandwidth and number of items to be migrated.



Data to migrate	Bandwidth	Time to migrate
1TB	1Gb/s (Internet)	140 minutes
1TB	10Gb/s (Internet)	14 minutes
1TB	100Gb/s (Express route)	1.4 minutes
1PB	1Gb/s (Internet)	100 days
1PB	10Gb/s (Internet)	10 days
1PB	100Gb/s (Express route)	1 day

» Cost. The cost for data migration may include the following:

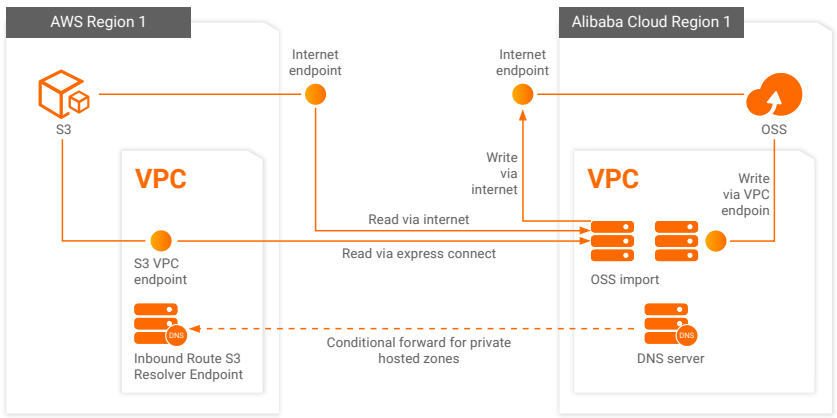
Fees	Description
Internet Egress fee from AWS	If you choose approach 1, AWS charges the internet egress fee.
Direct Connect Egress fee from AWS	If you choose approach 2, AWS charges the egress fee for direct connect. Typically, this fee is much lower than Internet Egress fee. Please refer to the pricing page.
API request fee from AWS	AWS charges API requests that will be used to retrieve data from S3

For example, the egress fee to the internet is as below. Please refer to this link [here](#) for more details.

Data Transfer OUT From Amazon S3 To Internet	
Up to 1 GB / Month	\$0.00 per GB
Next 9.999 TB / Month	\$0.09 per GB
Next 40 TB / Month	\$0.085 per GB
Next 100 TB / Month	\$0.07 per GB
Greater than 150 TB / Month	\$0.05 per GB

» Cross-region migration. If you plan to migrate to another region, it is recommended to use a close by region as an intermediate region and then do cross-region replication for the initial migration.

The high-level migration approach is illustrated below.



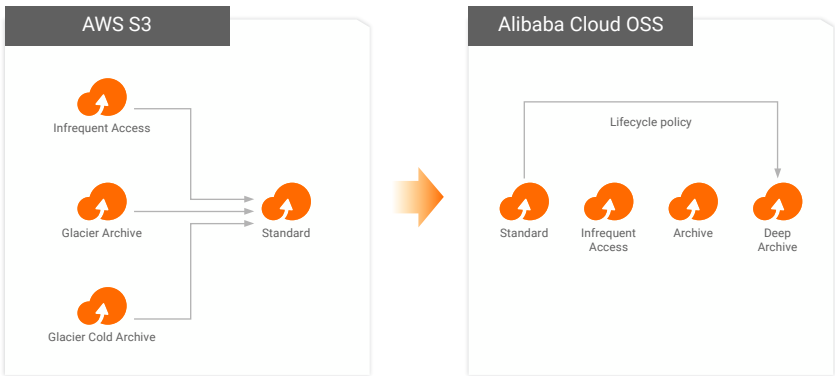
5. Migrating Data from S3 to OSS

Choose the right storage class

Below is a mapping between different storage classes of S3 and OSS. Choose the right storage class.

S3 Storage Class	OSS Cloud Storage Class
Standard	Standard
Infrequent Access	Infrequent Access
Archive (Glacier)	Archive
Deep Archive (Glacier)	Cold Archive

Since the Standard tier is optimized for frequent read and write, data migration should be done using this tier to optimize the migration cost.



Migration from S3 Standard to OSS via the Internet

The online migration feature in [Data Transport](#) is a serverless migration engine that natively allows you to migrate your AWS S3 buckets to OSS. Use this tool only when you do online migration via the Internet. If you use multi-version in S3, follow this [guide](#). The tool allows you to create S3 buckets and OSS as target addresses and then create migration jobs between the two targets.

Migration from S3 Standard to OSS via Express Connect

ossimport is a tool used to migrate data to OSS. You can deploy ossimport on local servers or ECS instances in the cloud to migrate data stored locally or in other cloud storage systems to OSS. It supports standalone mode (for <30TB migration) and distributed mode for (>=30TB migration). To be noted, when migrating ossimport read data from S3, you need to configure



conditional DNS forwarding for private hosted Zones. Refer to the "on-premises" section in this document for more details.

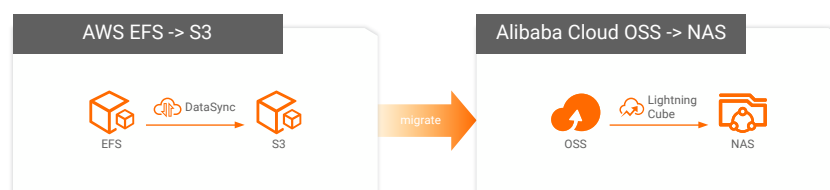
Migration from Infrequent Access, Archive, and Deep Archive to OSS

If you plan to migrate from Glacier Archive Tier or Glacier Deep Archive Tier to OSS Archive or OSS Cold Archive, since data retrieval from the archive tier incurs significant costs, it is recommended to convert to the Standard tier and follow the standard S3 to OSS approach for data migration.

6. Migration from EFS to NAS

Migration from EFS to NAS leverages S3 to OSS migration:

- » [Use DataSync to transfer data from EFS to S3](#)
- » Migrate S3 to OSS as described in the previous section
- » Use online migration to [transfer OSS to NAS](#)



4.4.2 APPLICATION MIGRATION

For object storage, it is recommended that you switch from S3 SDK to OSS SDK. While most of the APIs are compatible, there are a few differences. Refer to this document for [more details](#).

During the transition, your previous application might still be generating data and writing to S3. Your new application, based on OSS, needs to be able to read data that is newly generated. The back-to-origin setting allows you to retrieve the data from S3 when the data is not available in OSS. Follow this document for [more details](#).

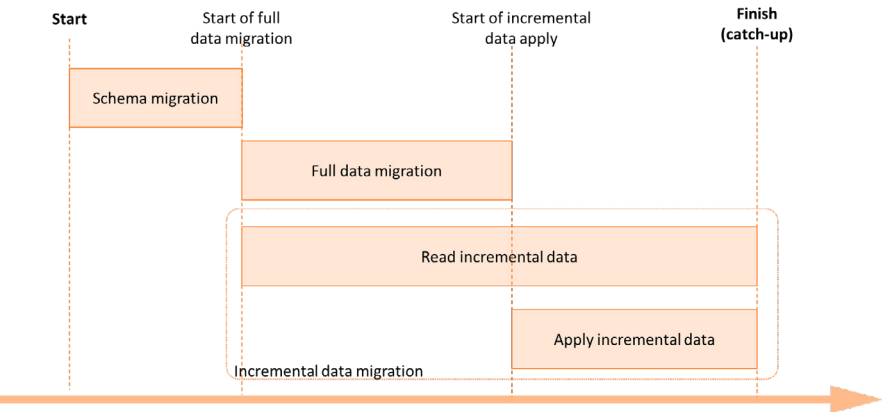
4.5 DATABASE

Database migration is always a cornerstone for cloud migration either from on-premise to public cloud or from one cloud platform to another. The database scheme is pretty clear and also features sophisticated migration techniques due to data reliability, high availability, application compatibility, and business continuity. For the overview of the database migration to Alibaba Cloud, please refer to https://www.alibabacloud.com/solutions/database_migration.



Specifically, we cover most cases when migrating databases from Amazon to Alibaba Cloud. We list the migration guide according to the types of databases below. Even for the same type of database, there may be more than one kind of alternative database cloud service as the target database on Alibaba Cloud.

Before heading into the database migration guide, let's talk more about [DTS](#), which is the key cloud service working as the data pipeline between the source and target databases, and works for [major migration scenarios](#). There are commonly three task types in DTS for migration:



Migration type	Description
Schema migration	DTS migrates the schemas of the required objects from the source database to the destination database. Tables, views, triggers, and stored procedures can be migrated. For schema migration between heterogeneous databases, DTS converts the schema syntax based on the syntax of the source and destination databases. For example, it converts the NUMBER data type in Oracle databases into the DECIMAL data type in MySQL databases.
Full data migration	DTS migrates historical data of the required objects from the source database to the destination database. If you select only Schema Migration and Full Data Migration as the migration types, DTS does not migrate incremental data generated in the source database. Note: To ensure data consistency, we recommend that you do not write data to the source database during full data migration. To ensure service continuity, you must select Schema Migration, Full Data Migration, and Incremental Data Migration when you configure a data migration task.





Migration type	Description
Incremental data migration	DTS retrieves static snapshots that are used for full data migration from the source database and migrates the snapshot data to the destination database. Then, DTS synchronizes incremental data generated in the source database to the destination database. Note: During incremental data migration, data is synchronized between the source and destination databases in real-time. The migration task does not automatically stop. You must manually stop the migration task.

4.5.1 GENERAL STEPS

Usually, there are applications running on top of the target database system on Amazon. Before we look at the migration actions, here are the general steps for most cases.

1. Target database planning and selection. Follow the questions to determine the target database on Alibaba Cloud:

- » Which database to use on Alibaba Cloud?
 - Cloud-hosted Alibaba Cloud Database vs Self-built database on ECS. Please refer to [ApsaraDB for RDS vs. Self-built Databases](#) for a brief comparison. Essentially, we recommend the [cloud-hosted Alibaba Cloud Database](#) for improved reliability, serviceability, cost-efficiency, upstream and downstream cloud service integration etc.
 - Alibaba Cloud Database Options. In case there are multiple options for a single type of database migration (e.g., RDS MySQL or PolarDB MySQL in section 2.4.1 MySQL migration, PolarDB Oracle, RDS MySQL, PolarDB MySQL, or AnalyticDB PostgreSQL in section 2.4.4 Oracle migration), we will elaborate the key difference in each section.
- » Which region, edition, and specification to use?
 - Please refer to the product purchase pages for database product availability in global regions.

Database Service	Global Region & Purchase Page
ApsaraDB RDS	https://rdsbuy.console.aliyun.com/create/rds/mysql
PolarDB	https://common-buy-intl.alibabacloud.com/?&commodityCode=polardb_sub_intl#/buy



Database Service	Global Region & Purchase Page
Redis	https://common-buy-intl.alibabacloud.com/?commodityCode=prepaid_kvstore_intl#/buy
MongoDB	https://common-buy-intl.alibabacloud.com/dds/prepay#/buy?
HBase	https://common-buy-intl.alibabacloud.com/?commodityCode=hbase_prepaid_public_intl
TSDB (Time Series Database)	https://common-buy-intl.alibabacloud.com/?commodityCode=hitsdb_pre_public_intl
InfluxDB	https://common-buy-intl.alibabacloud.com/?commodityCode=hitsdb_influxdbpre_public_intl
AnalyticDB MySQL	https://common-buy-intl.alibabacloud.com/?commodityCode=ads_pre_intl
AnalyticDB PostgreSQL	https://common-buy-intl.alibabacloud.com/?commodityCode=GreenplumPre_intl
ClickHouse	https://common-buy-intl.alibabacloud.com/?commodityCode=clickhouse_pre_public_intl

2. Network connectivity setup

- » DTS is the key service for the migration procedure. The network connectivity ensures that the DTS can connect to the target database on Amazon. DTS itself is a cloud service on Alibaba Cloud. After the target database has been determined and planned, it is recommended to create a DTS migration task at the same region with the target database on Alibaba Cloud (please refer to https://common-buy-intl.alibabacloud.com/?commodityCode=dtspost_intl to create DTS migration task in global regions).
- » It's important to make sure that the network connectivity from Alibaba Cloud to the source (Amazon) is working either through the public IP Port for database connection from DTS to the source database, or the network is connected via Cloud Enterprise Network (CEN), Express Connect, VPN Gateway, or Smart Access Gateway.
- » Please refer to <https://www.alibabacloud.com/help/doc-detail/84900.htm> to add the CIDR blocks of DTS servers at a specific region (where the DTS migration task is located) to the security settings of the source Amazon database.

3. Database migration

- » When the prerequisites are ready, the following sections will assist you with the actual migration process. Keep in mind



that the user cares about the connection status and data transmission performance during the DTS full data migration and incremental data migration. Please refer to the following documents for more information:

- [View the connection status and performance of full data migration](#)
- [View the connection status and performance of incremental data migration](#)

4. Application switch over

- » When the target database is running, the database connection Uniform Resource Identifier (URI) needs to be updated in the application to switch over to the new database on Alibaba Cloud.

4.5.2 MYSQL

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built MySQL on EC2	ApsaraDB RDS for MySQL This is the most common migration scenario with 100% MySQL compatibility.	https://www.alibabacloud.com/help/doc-detail/26621.htm
	PolarDB MySQL Migrate to cloud-native MySQL with computing and storage decoupled architecture to gain fast elasticity and up to 100TB data storage.	https://www.alibabacloud.com/help/doc-detail/100167.htm
Amazon RDS for MySQL	ApsaraDB RDS for MySQL This is the most common migration scenario.	https://www.alibabacloud.com/help/doc-detail/52555.htm
	PolarDB MySQL Migrate to cloud-native MySQL with computing and storage decoupled architecture to gain fast elasticity and up to 100TB data storage.	https://www.alibabacloud.com/help/doc-detail/100167.htm
Amazon Aurora MySQL	ApsaraDB RDS for MySQL When a user wants to migrate to the most commonly used RDS MySQL.	https://www.alibabacloud.com/help/doc-detail/112853.htm
	PolarDB MySQL This is the most common migration scenario.	https://www.alibabacloud.com/help/doc-detail/124691.htm

4.5.3 POSTGRESQL

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built PostgreSQL on EC2	ApsaraDB RDS for PostgreSQL	Full data migration: https://www.alibabacloud.com/help/doc-detail/120097.htm Incremental data migration for PostgreSQL versions 9.4.8 to 10.0: https://www.alibabacloud.com/help/doc-detail/26624.htm Incremental data migration for PostgreSQL versions 10.1 to 13: https://www.alibabacloud.com/help/doc-detail/147155.htm
Amazon RDS for PostgreSQL	ApsaraDB RDS for PostgreSQL	Full data migration: https://www.alibabacloud.com/help/doc-detail/112747.htm Incremental data migration: https://www.alibabacloud.com/help/doc-detail/142420.htm
Amazon Aurora PostgreSQL	ApsaraDB RDS for PostgreSQL	Full data migration: https://www.alibabacloud.com/help/doc-detail/112815.htm

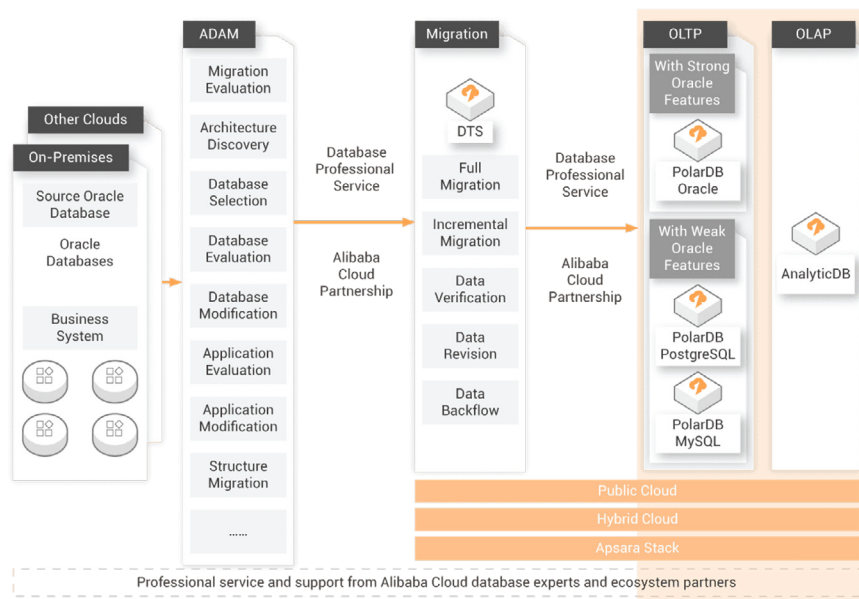
4.5.4 SQL SERVER

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built SQL Server on EC2	ApsaraDB RDS for SQL Server	Full data migration: https://www.alibabacloud.com/help/doc-detail/26622.htm Incremental data migration: https://www.alibabacloud.com/help/doc-detail/34357.htm
Amazon RDS for SQL Server	ApsaraDB RDS for SQL Server	Full data migration: https://www.alibabacloud.com/help/doc-detail/124678.htm

4.5.5 ORACLE

Migrating Oracle from Amazon to cloud-native databases on Alibaba Cloud would take more time. That's why a comprehensive [Oracle Migration](#) is provided by Alibaba Cloud.





Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built Oracle on EC2	PolarDB Oracle We recommend this migration option with cloud-native features: high compatibility with Oracle but with cost reduction	https://www.alibabacloud.com/help/doc-detail/144033.htm
	ApsaraDB RDS for MySQL If you decide to migrate to the MySQL ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	https://www.alibabacloud.com/help/doc-detail/26620.htm
	PolarDB MySQL If you decide to come to MySQL ecosystem (with cloud-native MySQL on Alibaba Cloud) with strategic consideration in the long term, this can avoid vendor lock-in.	https://www.alibabacloud.com/help/doc-detail/183514.htm
	AnalyticDB PostgreSQL Migrate from Oracle RAC (Real Application Clusters) or Oracle data warehouse in most cases.	Data migration: https://www.alibabacloud.com/help/doc-detail/171699.htm Application migration: https://www.alibabacloud.com/help/doc-detail/124685.htm
	Message Queue Instance: Alibaba Cloud Kafka Connect to Oracle on Amazon and get full data and incremental data to the message queue. The downstream application or data system then consumes message data from Kafka.	https://www.alibabacloud.com/help/doc-detail/207737.htm

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Amazon RDS for Oracle	ApsaraDB RDS for MySQL If you decide to come to the MySQL ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	https://www.alibabacloud.com/help/doc-detail/120156.htm
	PolarDB Oracle We recommend this migration option with cloud-native features, which have high compatibility with Oracle but with cost reduction.	Please configure to open the public endpoint for Amazon RDS for Oracle, before following this migration guide: https://www.alibabacloud.com/help/doc-detail/144033.htm

4.5.6 IBM DB2

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built IBM Db2 on EC2	ApsaraDB RDS for MySQL If you decide to come to the MySQL ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	https://www.alibabacloud.com/help/doc-detail/74779.htm

4.5.7 MONGODB ATLAS

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
MongoDB Atlas on Amazon	ApsaraDB for MongoDB	Migration using DTS: https://www.alibabacloud.com/help/doc-detail/142448.htm Migrate data from MongoDB Atlas to ApsaraDB for MongoDB by using mongodump and mongorestore: https://www.alibabacloud.com/help/doc-detail/102305.htm



4.5.8 DYNAMODB

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Amazon DynamoDB	ApsaraDB for MongoDB If you decide to come to the MongoDB ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	Migrate an Amazon DynamoDB database to ApsaraDB for MongoDB by using NimoShake: https://www.alibabacloud.com/help/doc-detail/172828.htm Migrate data from Amazon DynamoDB to ApsaraDB for MongoDB by using mongoimport: https://www.alibabacloud.com/help/doc-detail/115570.htm Use NimoFullCheck to check data consistency after migration: https://www.alibabacloud.com/help/doc-detail/172827.htm

4.5.9 ELASTICACHE/REDIS

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Amazon ElastiCache	ApsaraDB for Redis	https://www.alibabacloud.com/help/doc-detail/101071.htm
Self-built Redis on EC2	ApsaraDB for Redis	Migration using DTS: https://www.alibabacloud.com/help/doc-detail/183620.htm Migration using Redis-Shake (within the guide, perform a similar operation on Amazon EC2 just like on Alibaba Cloud ECS): https://www.alibabacloud.com/help/doc-detail/111066.htm

4.5.10 REDSHIFT

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Amazon Redshift	AnalyticDB PostgreSQL	https://www.alibabacloud.com/help/doc-detail/101215.htm

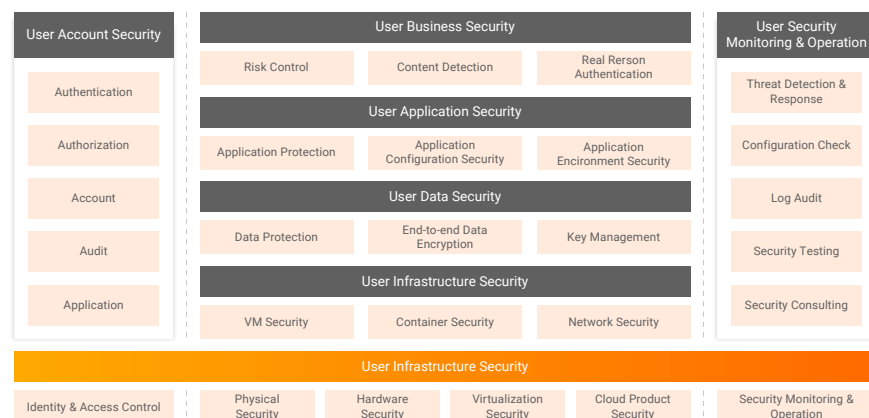


4.5.11 GREENPLUM

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built Greenplum on EC2	AnalyticDB PostgreSQL	https://www.alibabacloud.com/help/doc-detail/157271.htm

4.6 SECURITY

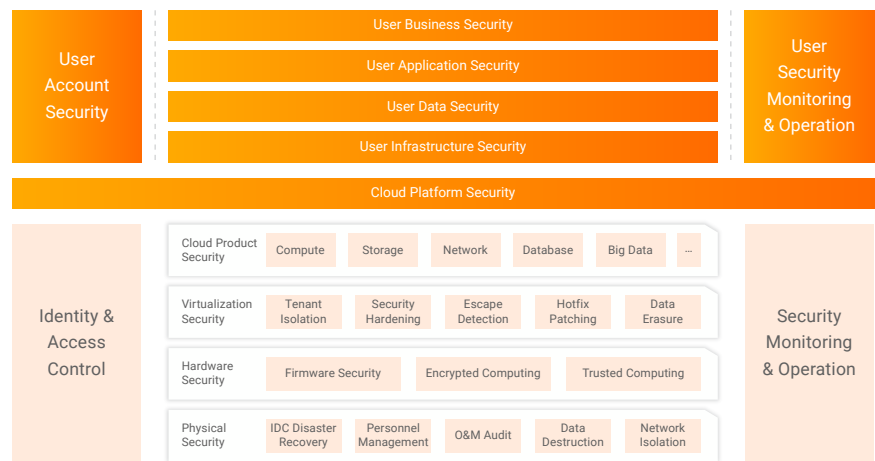
4.6.1 ALIBABA CLOUD SECURITY ARCHITECTURE



Alibaba Cloud provides a security architecture that offers protection in five horizontal dimensions and two vertical dimensions. The two vertical dimensions include account security (identity and access control) and security monitoring and operations. Note that these vertical dimensions include different implementations for both user security and cloud platform security. The five horizontal dimensions includes cloud platform security at the bottom layer, as well as infrastructure security, data security, application security, and business security for cloud users.

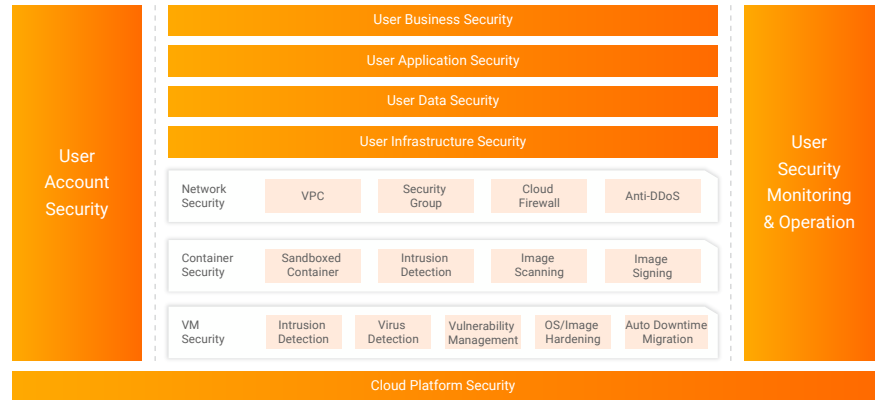
This chapter introduces the overall security architecture and describes the key features of each layer that covers various Alibaba Cloud products and security services.

Cloud platform security



The cloud platform security architecture includes the basic security capabilities provided by the Alibaba Cloud platform. In the two vertical dimensions, ‘Identity & Access Control’ and ‘Security Monitoring & Operations’ are invisible to customers. Similarly, customers can enjoy high-level protection capabilities of Alibaba Cloud without performing configurations at the physical security, hardware security, and virtualization security layers. The cloud product security layer includes the security capabilities provided by the cloud platform for customers. Some capabilities such as tenant isolation are enabled by default. Whereas other capabilities, such as data encryption, must be enabled and configured by customers to function properly.

User infrastructure security



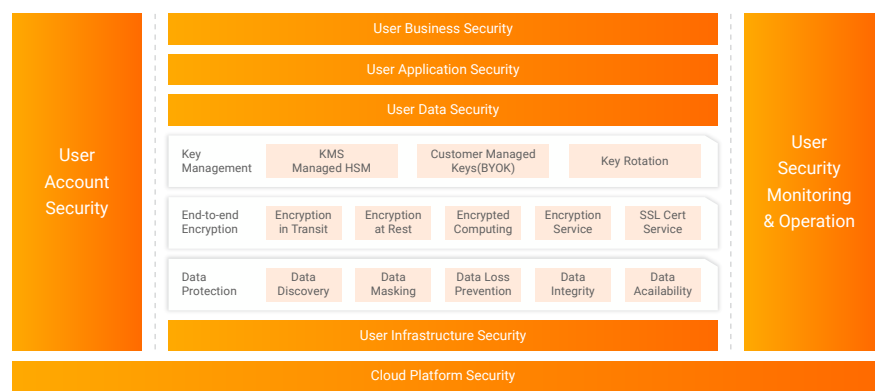
The user infrastructure security capabilities and requirements of cloud users focus on three aspects: host security, container security, and network security. These aspects involve the use of networks to isolate and protect the most important computing resources. Cloud applications and services of users are directly or indirectly built through upper-layer cloud services and run on the basic computing and network service modules.





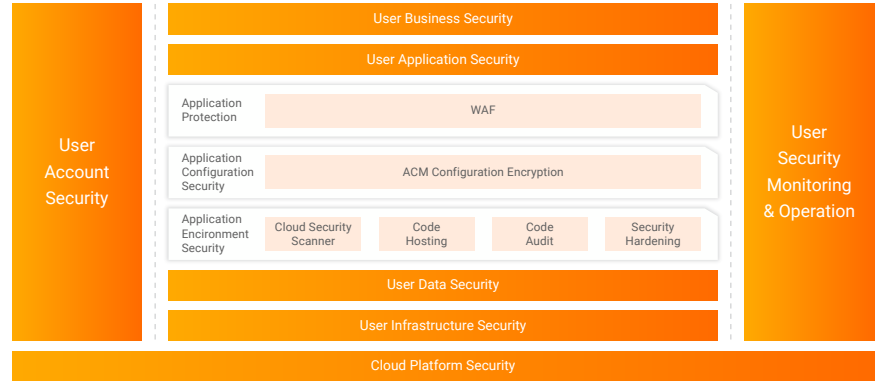
In addition to computing and network resources, storage resources are also crucial to users, especially in the protection of user data. The user-side security architecture also includes two vertical dimensions: user account security and user security monitoring and operations. These vertical dimensions are involved in each horizontal dimension, such as the configuration security checks for VM hosts during security monitoring and control, over permissions of various cloud resources for account security.

User data security



User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal states that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belongs only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have the responsibility and obligation to help users ensure the confidentiality, integrity, and availability of their data.

User application security



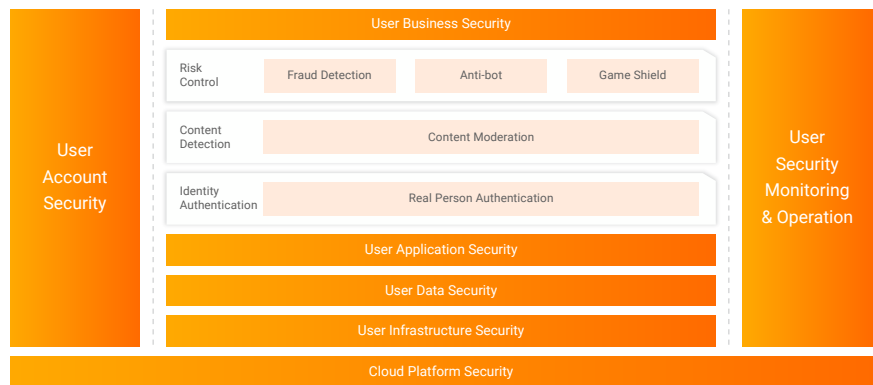
Applications built on Alibaba Cloud also need to be properly secured. At the application security level, Alibaba Cloud provides users with security capabilities in three aspects: application environment



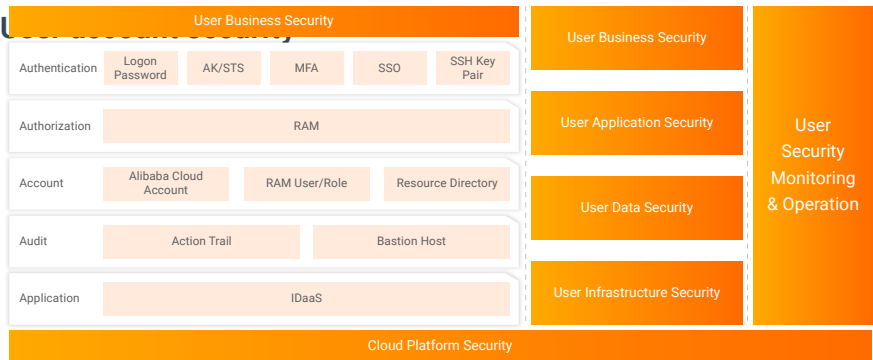


security, application configuration security, and application protection.

User business security



After a user's application has been built and is running on the Alibaba Cloud platform, Alibaba Cloud provides corresponding security capabilities for different business scenarios of the user. At the business security level, Alibaba Cloud provides security features from three aspects: identity authentication, content detection, and risk control.

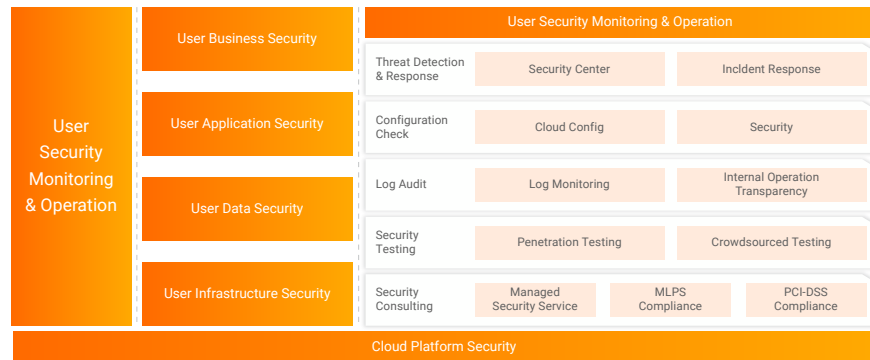


User account security is an important dimension in the overall cloud security architecture design. User account security on the cloud involves the “Five A’s”: Authentication, Authorization, Account management, Audit, and Application management. It is important to note that within a cloud-based business, there is often a cloud product or service (such as RAM) that actually provides multiple dimensions of capabilities in the aforementioned aspects.

This chapter describes Alibaba Cloud account security capabilities specific to each dimension.



User security monitoring and operations



Once a user's resources, applications, business, and accounts are properly protected on the cloud, the cloud platform needs to provide security monitoring and operations capabilities to detect security threats, monitor the configurations of cloud resources and logs, and (where appropriate) perform security testing on the user's overall business environment in order to provide a response and consulting services. In this way, the security monitoring and operations capabilities of Alibaba Cloud can effectively safeguard the user's business.

4.6.2 ANTI-DDOS

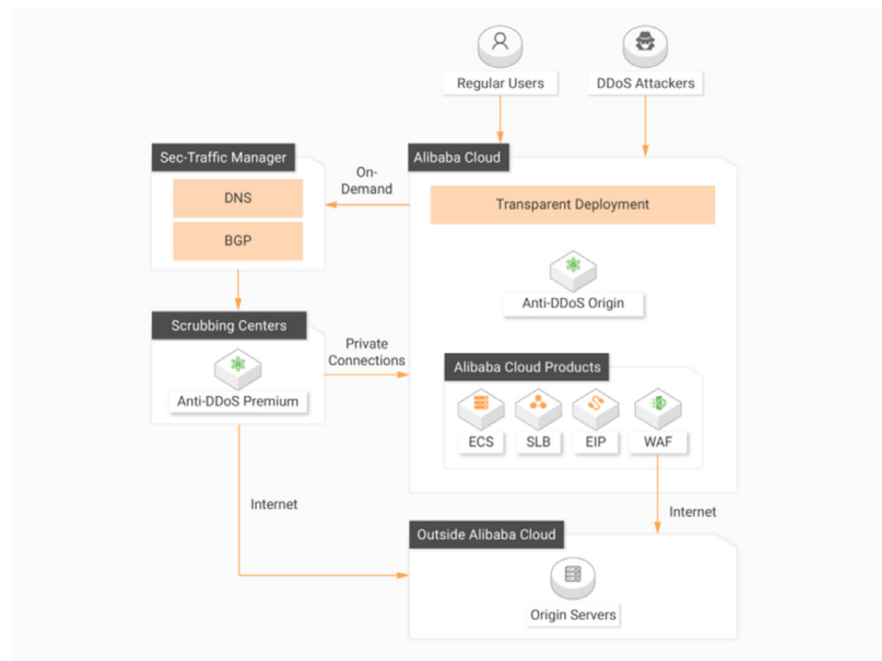
Anti-DDoS is provided for enterprises whose services on the Internet become unavailable due to distributed denial of service (DDoS) attacks.

By default, Anti-DDoS provides capacity of up to 5 Gbit/s for free. In addition, Alibaba Cloud has launched the Security Credibility plan. After becoming a member of Security Credibility, you can enjoy the additional DDoS mitigation capacity on top of the default offering based on your security credibility score.

DDoS protection package

The Anti-DDoS protection package is designed for large-scale enterprises on Alibaba Cloud. It uses the native Alibaba Cloud network and transparent protection engine to mitigate DDoS attacks. The protection package can be used to protect various Alibaba Cloud products, such as Elastic Compute Service (ECS), Server Load Balancer (SLB), Web Application Firewall (WAF), and Elastic IP Address (EIP).





» Procedure

» Step 1: Purchase an Anti-DDoS Origin Enterprise instance



NOTE

Anti-DDoS Origin Enterprise instances are only available in Mainland China. If you want to purchase an Anti-DDoS Origin Enterprise instance outside of Mainland China, you must submit a ticket or contact sales personnel. After the request is approved, you will be able to purchase an Anti-DDoS Origin Enterprise instance.

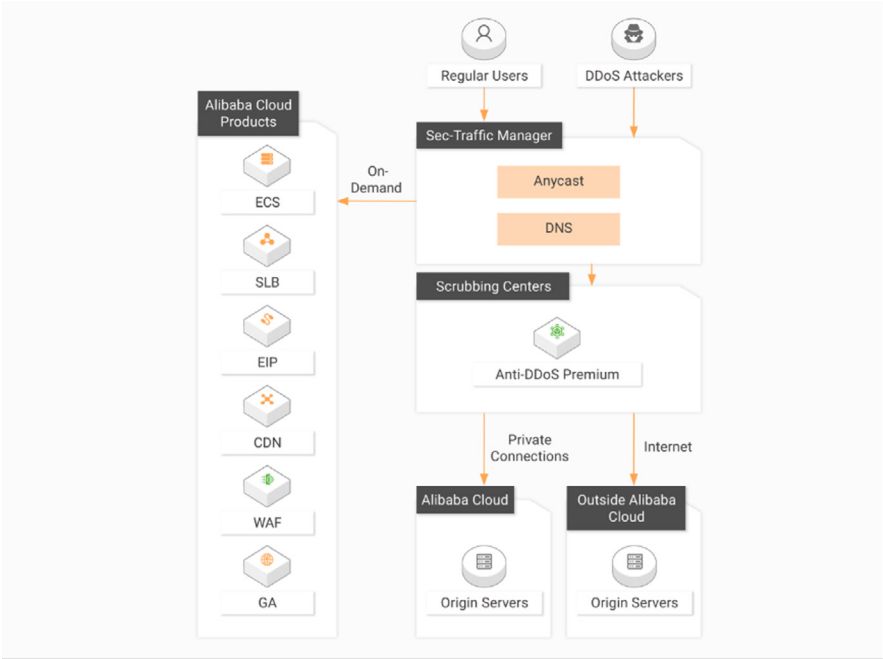
» Step 2: Log on to the Anti-DDoS Basic console.

» Step 3: On the Anti-DDoS Origin page, locate the target instance and click 'Add a protection target' to add the addresses that you want to protect to the instance.

Anti-DDoS Pro and Premium

Anti-DDoS Pro and Anti-DDoS Premium are designed for enterprises that deploy their resources on or off of Alibaba Cloud. By using the massive traffic scrubbing center resources of Alibaba Cloud, both products work together with the Artificial Intelligence (AI) protection engine and adopt full-traffic proxy to protect against high-volume traffic attacks and refined web application level resource exhaustion attacks, such as HTTP (HyperText Transfer Protocol) flood attacks.





Set up Anti-DDoS Pro using domains

Step 1: Add forwarding rules	In the Anti-DDoS Pro or Anti-DDoS Premium console, you can add a website service that you want to protect by using a domain name, associate the service with an Anti-DDoS Pro or Anti-DDoS Premium instance, and configure the traffic forwarding rules.
Step 2: Configure service traffic forwarding	Modify the DNS records of your domain name to reroute the traffic directed to your website to an Anti-DDoS Pro or Anti-DDoS Premium instance. The instance scrubs the traffic and then forwards the traffic to the origin server, which protects your website service against DDoS attacks.
Step 3: Configure protection policies	After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, Intelligent Protection is automatically enabled. You can manually adjust anti-DDoS protection policies to suit your website service, which includes Intelligent Protection, Black Lists and White Lists (Domain Names), Blocked Regions (Domain Names), Accurate Access Control, and Frequency Control.
Step 4: View the protection data of your website services	After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, you can use the Security Reports feature and log-related features to view the protection data in the Anti-DDoS Pro or Anti-DDoS Premium console.

Set up Anti-DDoS Pro using IPs and ports

Step 1: Create a port forwarding rule	Add a non-website service that you want to protect by using a port in the Anti-DDoS Pro or Anti-DDoS Premium console. Use the IP address of your Anti-DDoS Pro or Anti-DDoS Premium instance as your service IP address to reroute inbound traffic to your instance. After you change the IP address, the instance scrubs the inbound traffic and then forwards the traffic to the origin server.
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Step 2: Configure port forwarding and anti-DDoS protection policies

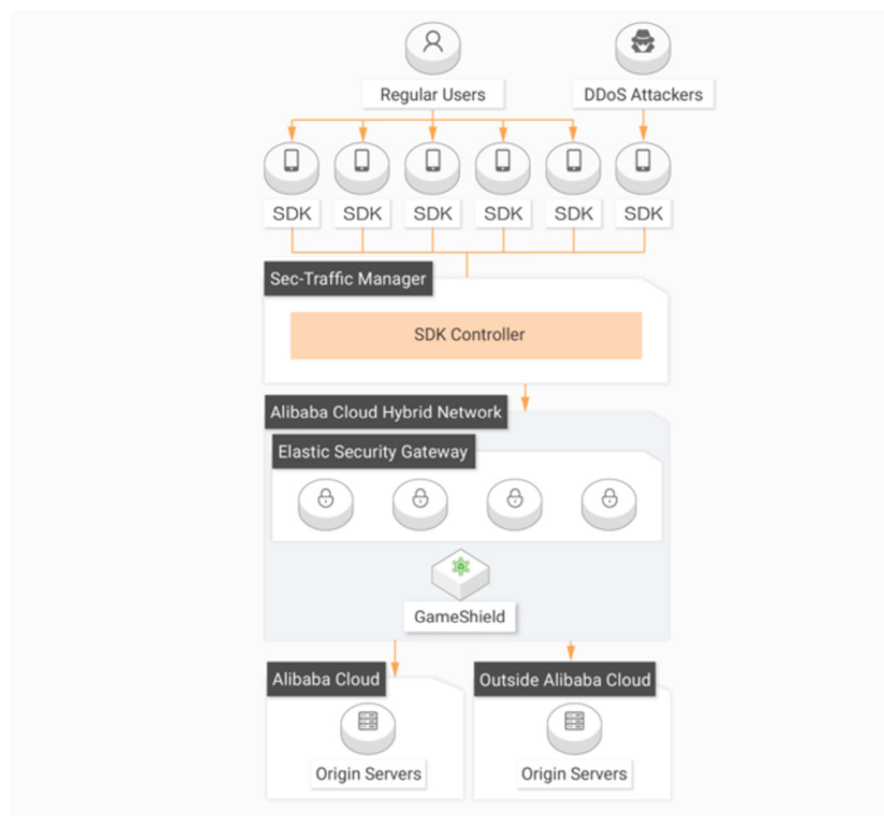
Configure port forwarding policies as required, such as session persistence and health checks for multiple origin IP addresses. You can also configure anti-DDoS protection policies for non-website services, such as False Source, Speed Limit for Destination, Packet Length Limit, and Speed Limit for Source.

Step 3: View the protection data of a port

View the traffic that goes through a port on the Security Overview page of the Anti-DDoS Pro or Anti-DDoS Premium console after you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your non-website service.

GameShield

GameShield is a network security solution presented by Alibaba Cloud for protecting the gaming industry against DDoS attacks. Compared with Anti-DDoS Pro, GameShield can defend against high-traffic DDoS attacks at the Tbps (Terabytes Per Second) level and TCP (Transmission Control Protocol) resource exhaustion attacks (Layer-4 CC attacks) that are especially applicable to the gaming industry. This solution reduces protection costs and provides better protection.



» Procedure

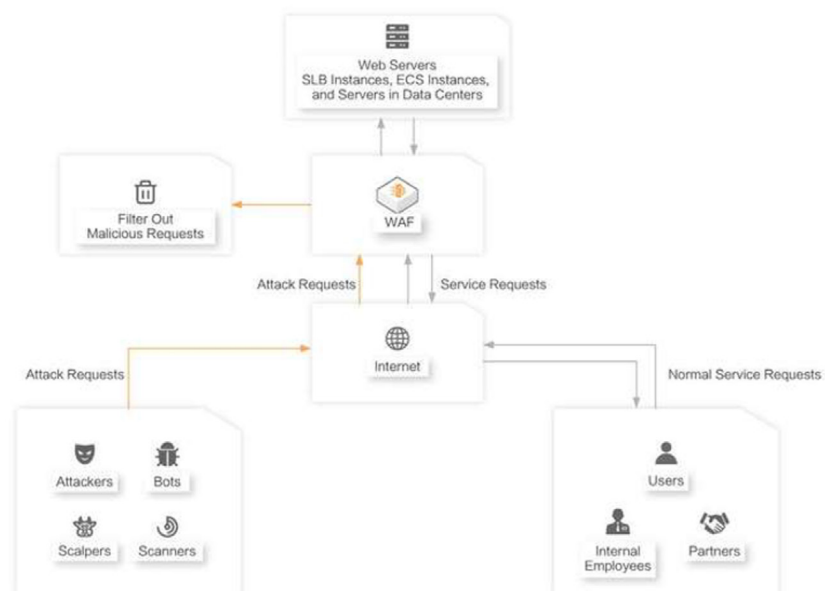
- » **Step 1:** Create a game instance.
- » **Step 2:** Configure a protection object.
- » **Step 3:** Add your application to GameShield.



4.6.3 WAF

Based on the big data and intelligent computing capabilities of Alibaba Cloud Security, Alibaba Cloud WAF defends against common security threats reported by OWASP (Open Web Application Security Project). These threats include SQL injection, cross-site scripting (XSS), common vulnerabilities in web server plugins, web shell upload, and unauthorized access to cloud resources. WAF also filters out vast numbers of malicious access attempts, prevents leakage of web assets and data, and ensures the security and availability of web applications.

When you add your website to WAF in CNAME (Canonical Name) record mode, you must enter the website information and change the DNS record to resolve the domain name of your website to the CNAME assigned by WAF. This way, the requests destined for your website are redirected to WAF. This mode is supported regardless of whether your origin server is deployed on the cloud.

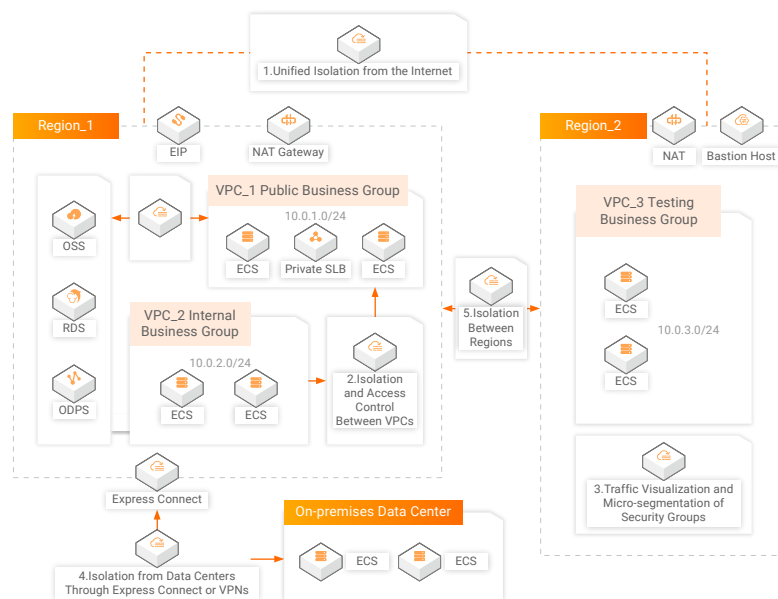


Alibaba Cloud offers Hybrid Cloud WAF to help protect and manage web applications in various hybrid cloud environments in a centralized manner. The web applications can be deployed across Alibaba Cloud Public Cloud and data centers, as well as across Alibaba Cloud Public Cloud and third-party clouds. In this case, Hybrid Cloud WAF protects your service traffic, which cannot be migrated to the cloud. To deliver an elastic and effective system, Hybrid Cloud WAF combines shared and exclusive resources in and outside the cloud. This way, you can protect the web applications in a centralized manner.



4.6.4 CLOUD FIREWALL

Cloud Firewall is the industry's first firewall as a service (FWaaS) solution for public cloud environments. This product manages the access control policy for North-South Traffic from the Internet to ECS instances and the micro-isolation policy for East-West Traffic between ECS instances in a centralized manner. Cloud Firewall comprehensively checks the exposure of your assets to the Internet in the cloud environment and manages the access policies for public network IP addresses in a centralized manner. With one-click access, Cloud Firewall is the primary network security infrastructure for ensuring the network security of your business(es) migrated to Alibaba Cloud. With the built-in intrusion prevention system (IPS), this product provides intelligent prevention, detects victim hosts, blocks external connections launched by hosts, and visualizes network-wide traffic and access relationships among ECS instances. Cloud Firewall meets the requirements of MLPS 2.0 (Multi-Level Protection Scheme version 2.0) for virtual boundaries, access control, IPS, and storage time of security logs, which must be retained for at least six months. This product is a necessary choice to be compliant with MLPS 2.0.



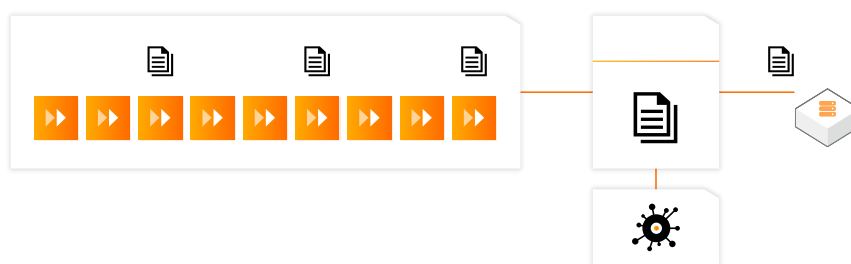
» Procedure

- » **Step 1:** On the Firewall Settings page, enable firewalls.
- » **Step 2:** Choose Intrusion Prevention > Intrusion Prevention to configure intrusion prevention policies.
- » **Step 3:** On the Access Control page, configure access control policies.
- » **Step 4:** On the Traffic Analysis page, view traffic statistics. You can also view information about outbound connections.



4.6.5 SECURITY CENTER

Security Center is a unified security management system that identifies, analyzes, and alerts security threats in real-time. With security capabilities such as anti-ransomware, anti-virus, tamper-proofing, and compliance assessment, users can automate security operations, response, and threat tracing to safeguard cloud and local servers, and meet regulatory compliance requirements.



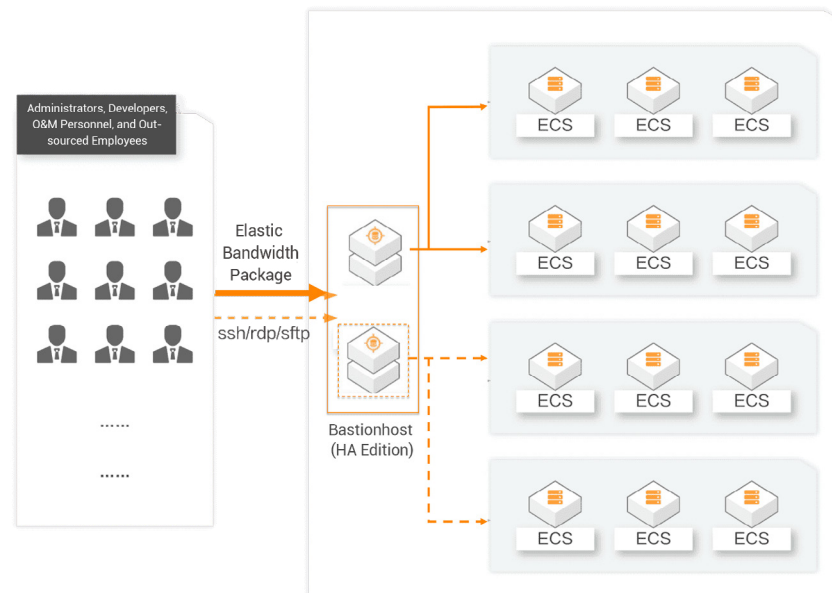
» Procedure

- » **Step 1:** Log on to the Security Center console.
- » **Step 2:** In the Unprotected assets (ECS) section on the Overview tab, you can view the number of unprotected servers that do not have the Security Center agent installed.
- » **Step 3:** Click Install now to go to the Settings page. Click the Agent tab and then the Client that is to be installed tab. On the tab that appears, you can view the total number and list of servers that do not have the Security Center agent installed.
- » **Step 4:** Click the Client Installation Guide tab to choose automatic installation or, alternatively, download the installation package and run commands to install the Security Center agent.

4.6.6 BASTION HOST

Bastion Host provides multiple functions, such as O&M identity authentication, account management, and system operation audit. Bastion Host acts as a forward proxy and records the data flow of common O&M protocols, such as Secure Shell (SSH), Remote Desktop Protocol (RDP), and Secure File Transfer Protocol (SFTP). It can also regroup the data flows and reproduce them to implement an O&M audit.





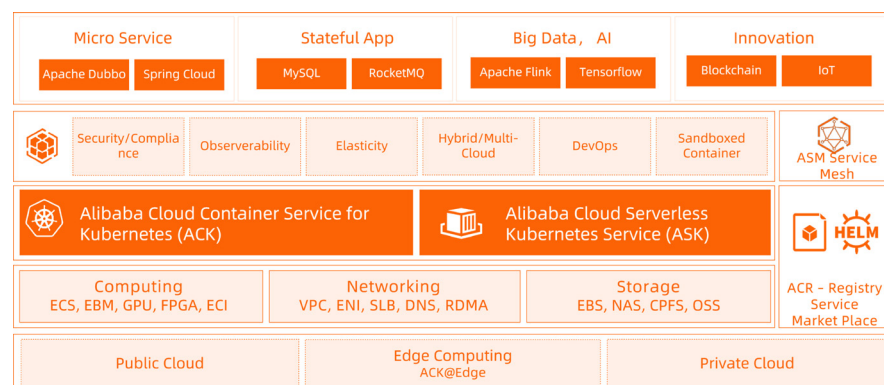
» Procedure

- » **Step 1:** Synchronize ECS assets.
- » **Step 2:** Import Alibaba Cloud RAM users.
- » **Step 3:** Create O&M rules.
- » **Step 4:** Perform O&M operations on hosts.
- » **Step 5:** Audit O&M sessions.

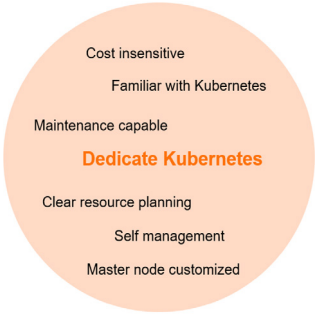
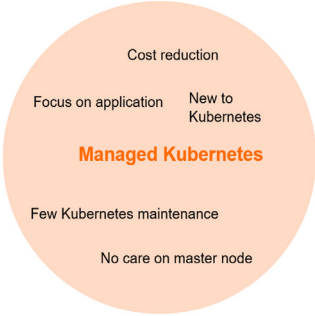
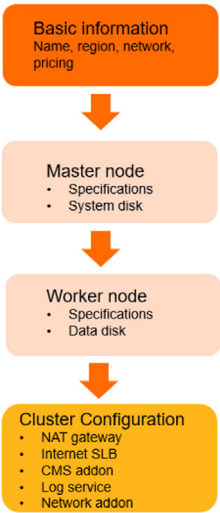
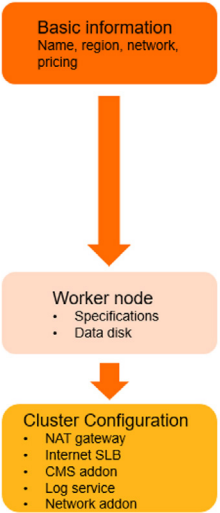
4.7 CONTAINER / MIDDLEWARE

Container Service for Kubernetes (ACK) is one of the first services to pass the Certified Kubernetes Conformance Program. ACK provides high-performance management services for containerized applications. You can manage enterprise-level containerized applications throughout the application lifecycle.

The following figure shows the architecture of the Alibaba Cloud Container Service product portfolio.

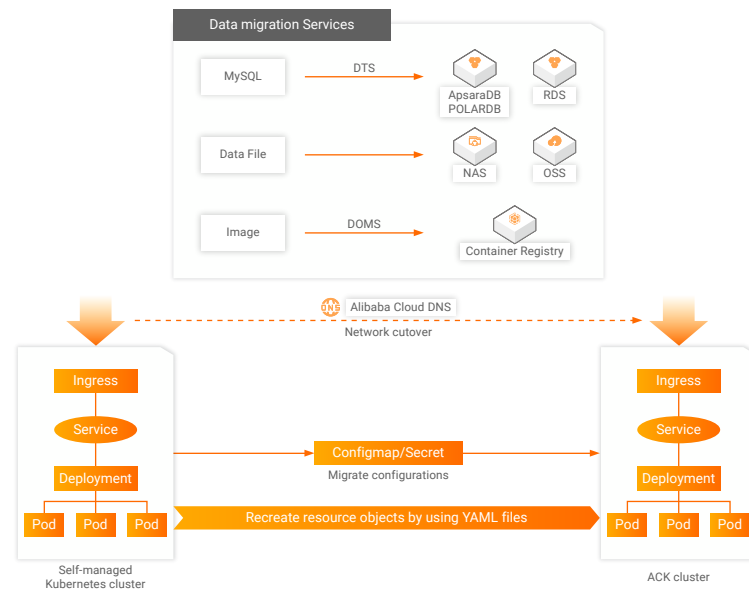


ACK provides the following three cluster types: dedicated Kubernetes clusters, managed Kubernetes clusters, and serverless Kubernetes clusters.

Item	Dedicated Kubernetes cluster	Managed Kubernetes cluster	Serverless Kubernetes
Feature	You must create master nodes and worker nodes. A dedicated Kubernetes cluster allows you to manage the cluster infrastructure in a more fine-grained manner. You must design, maintain, and upgrade the Kubernetes cluster on your own.	You only need to create worker nodes. ACK creates and manages master nodes. A managed Kubernetes cluster is easy-to-use, cost-efficient, and highly available. You do not need to manage master nodes.	You do not need to create master or worker nodes. A serverless Kubernetes cluster allows you to start applications directly. You do not need to manage nodes.
Billing method	You are not billed for cluster management features. However, you must pay for master nodes, worker nodes, and other infrastructure.	» Standard ACK clusters: You are not billed for cluster management features. However, you must pay for worker nodes and other infrastructure. » Professional ACK clusters: You are billed based on the number of clusters or the subscription method.	You are billed for the resources consumed by pods and the resource usage duration. The duration is measured in seconds.
Use scenario	Applies to all scenarios.	Applies to all scenarios.	Applies to batch tasks, urgent application scale-out, and continuous integration (CI) or continuous delivery (CD).
User profile	 Dedicated Kubernetes	 Managed Kubernetes	 Serverless Kubernetes
Cluster creation procedure			

4.7.1 HOW TO MIGRATE AN AWS EKS MANAGED KUBERNETES CLUSTER TO CONTAINER SERVICE FOR KUBERNETES (ACK) WITHOUT SERVICE DISRUPTION.

Migration scheme



Procedure

1. **Create and configure an ACK cluster.** O&M engineers create an ACK cluster and configure cluster resources. This step makes it easy for developers to migrate applications to Kubernetes. For more information, see [Create a dedicated Kubernetes cluster](#). The following parameters are required when you create the ACK cluster:

- » **Cluster type:** Select one of the following cluster types when you create the ACK cluster:
 - Standard dedicated Kubernetes cluster. You need to create and manage the master and worker nodes of the ACK cluster. You have full control over the ACK cluster and are charged for the resources that are used by the master and worker nodes.
 - Standard managed Kubernetes cluster. You need to create and manage only the worker nodes of the ACK cluster. ACK creates and manages the master nodes, and you are only charged for the resources that are used by the worker nodes.
- » **Operating system:** Select an operating system based on your requirements:
 - By default, the CentOS 7.6 or Aliyun Linux 2.1903 operating system is selected. We recommend that you keep the default settings.

- If the default system kernel cannot meet your requirements, you can use a custom image. For more information, see [Use a custom image to create an ACK cluster](#).
- » **VPC:** Select a virtual private cloud (VPC) and vSwitches for the ACK cluster.
- » **SNAT:** Configure Source Network Address Translation (SNAT) entries for the VPC.
- » **Public access:** Expose the API server of the ACK cluster with an Elastic IP address (EIP).
- » **Cloud Monitor agent:** Install the Cloud Monitor agent on the Elastic Compute Service (ECS) instances of the ACK cluster.
- » **Log Service:** Install and configure the Log Service agent in the ACK cluster.

2. Migrate data.

- » Migrate databases:
 - i. Create a Relational Data Service (RDS) instance.
 - ii. Configure an RDS whitelist to allow only specified IP addresses to access the RDS instance.
 - iii. Configure a PrivateZone. Use the PrivateZone to resolve the domain name of the database to the IP address of the RDS instance. This saves you the need to modify the database configuration on applications.
 - iv. Migrate the data in the MySQL database of the self-managed Kubernetes cluster. Use Data Transmission Service (DTS) to migrate the data from the MySQL database to the ApsaraDB for RDS database in full, incremental, or two-way synchronization mode. For more information, see [Migrate data from a user-created MySQL database to an ApsaraDB RDS for MySQL instance](#).
- » Migrate data.
 - i. Activate Object Storage Service (OSS).
 - ii. Create an OSS bucket.
 - iii. Migrate the data that is stored in the self-managed Kubernetes cluster. Use the ossimport tool to migrate the data in batches from an on-premises server or a third-party cloud storage service to OSS. For example, you can choose a third-party cloud storage service such as Amazon S3, Microsoft Azure, or Tencent Cloud Object



Storage. For more information, see [Architectures and configurations](#).

» Migrate images:

- i. Create a Container Registry repository.
- ii. Set the credential that is used to access the created Container Registry repository.
- iii. Migrate the images of the self-managed Kubernetes cluster. You can use the image-syncer tool to migrate the images of the self-managed Kubernetes cluster to the Container Registry repository. For more information, see [Use image-syncer to migrate container images](#).

- 3. Migrate application configurations.** Container Service for Kubernetes (ACK) allows you to back up and restore stateful applications deployed in an ACK cluster. This is an all-in-one solution to achieve crash consistency, application consistency, and cross-region disaster recovery for stateful applications in ACK clusters. For more information, see [Migrate applications to an ACK cluster](#).

Prerequisites: Two Kubernetes clusters of the same version are created. For more information, see [Install the application backup component](#).

Method 1: Migrate an application by using the application backup feature

- a. Grant the required Object Storage Service (OSS) permissions to Cluster_A and Cluster_B. For more information, see [Configure OSS permissions](#).
- b. Install the application backup component in Cluster_A and Cluster_B. For more information, see [Install the application backup component](#).
- c. Create backup vaults for Cluster_A and Cluster_B. When you create backup vaults for these, select the same OSS bucket and set Network Type to Public Network. For more information, see [Create a backup vault](#).
- d. Create a backup task in Cluster_A. For more information, see [Create a backup task](#). After you create the backup task in Cluster_A, you can log on to Cluster_B to view the backup task on the **Backups** tab of the **Application Backup** page.
- e. On the **Application Backup** page, click the **Restore** tab and create a restoration task based on the backup task that is



created in Cluster_A. For more information, see [Create a restoration task](#).

Method 2: Migrate an application by using Velero

This example shows how to migrate an application in the Nginx-examples namespace from Cluster_A to Cluster_B. To migrate the application, perform the following steps:

- a.** Install the Velero client and the application backup component in Cluster_A. Then, configure and connect to the OSS bucket named 'mybackups'. For more information, see [Velero](#) and [Install the application backup component](#).
 - b.** Install the Velero client and migrate-controller in Cluster_B. Then, configure and connect to the OSS bucket named **mybackups**.
 - c.** Back up applications in the **nginx-examples** namespace in Cluster_A and specify the name of the backup file, for example, **nginx-backup**.
 - d.** Use the backup file named **nginx-backup** in Cluster_B to restore the applications. This operation migrates the applications from Cluster_A to Cluster_B.
- 4. Perform regression tests.** Test engineers perform regression tests on the ACK cluster without interrupting the online business:
 - » Configure the domain names for regression tests.
 - » Test applications.
 - » Check the logged application events.
 - » Check the monitoring metrics of the applications.
- 5. Switch all production traffic to ACK.** O&M engineers modify the Domain Name System (DNS) configuration to switch traffic to the ACK cluster:
 - » Use the DNS service: Modify the DNS configuration to switch the traffic.
 - » Upgrade the configurations or code of the client to switch the traffic.
- 6. Bring the self-managed Kubernetes cluster offline.** O&M engineers check whether the ACK cluster can be accessed as expected. Then, O&M engineers bring the self-managed Kubernetes cluster offline:
 - » Check whether the ACK cluster can receive and send traffic as expected.
 - » Bring the self-managed Kubernetes cluster offline.



- » Clear the backup files stored in the OSS bucket that is created when you migrate applications.

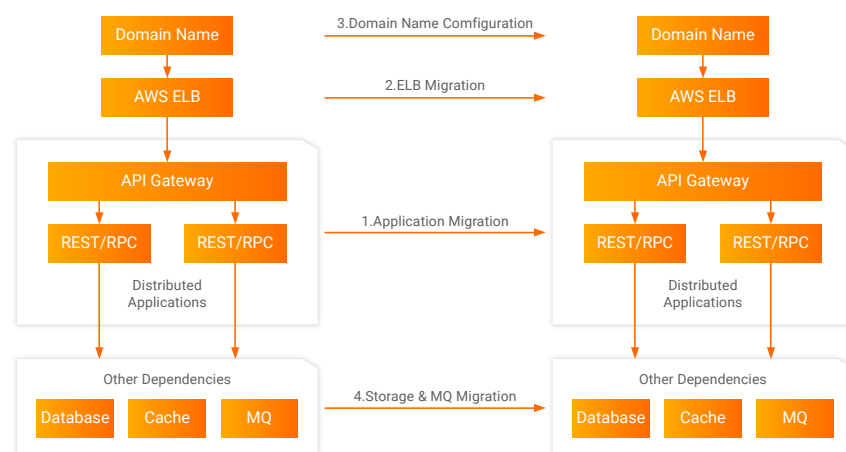
4.8 MICROSERVICES

Enterprise Distributed Application Service (EDAS) is an Application Platform as a Service (aPaaS) offering that allows you to host applications and manage microservices. EDAS provides full-stack solutions to develop, deploy, monitor, and maintain applications, as well as support microservice frameworks such as Dubbo and Spring Cloud, which is a good choice to migrate microservice-based applications from AWS to Alibaba Cloud.

Since your applications have been deployed to the production environment and are running in a normal state, it is the first priority to ensure the smooth operation of the business without interruption during the migration process. At this point, Alibaba Cloud has very good support in this scenario, which can ensure that the application platform runs uninterrupted during the migration process.

4.8.1 MIGRATION PROCEDURE

Specifically, Alibaba Cloud supports two kinds of application migration of Spring Cloud and Dubbo microservice framework. The migration process of these two types of applications are the same, mainly including the following four aspect:



1. Application Migration

Applications to be migrated are typically stateless and can be migrated first. For more details on this, please refer to the following documents:

- » Spring Cloud:

<https://www.alibabacloud.com/help/doc-detail/120424>.

[htm#title-x9c-woq-wdi](#)

» Dubbo:

<https://www.alibabacloud.com/help/doc-detail/108540.htm#title-mpb-nxa-812>

2. ELB Migration

- » If the AWS ELB instance is used before migration, then after the application migration is complete, you need to migrate from ELB to SLB instance on Alibaba Cloud.
- » If no AWS ELB instance is used before migration, it's recommended that you create and bind an SLB instance to the application after migration.

3. Domain Name Configuration

To create and bind a new SLB instance on Alibaba Cloud to migrated applications, you must add this SLB instance configuration to the domain name configuration. For more information, see [Change DNS servers for a domain name](#).

4. Storage and MQ Migration (Optional)

If your applications were deployed on AWS Elastic Beanstalk using other AWS services, such as Amazon RDS and Amazon MQ, you still need to migrate storage and message queues along with applications. For this part, you can follow [Chapter 2.4](#) database migration.

4.8.2 MIGRATION SOLUTION

You can migrate applications by using two methods: registry switching, and dual registration and dual subscription. The two methods allow you to migrate applications during runtime without service discontinuity.

Microservice frameworks on Amazon	Migration Solution	Description
Spring Cloud	Registry switching	Use Spring Cloud Alibaba to switch the original service registry to Nacos. Develop applications and deploy them to EDAS, and switch traffic by configuring the SLB and the domain name. Refer to this document for more details .
	Dual registration and dual subscription	Access the original service registry and the EDAS service registry when applications are migrated to enable mutual calling between migrated applications and non-migrated applications. Refer to this document for more details .



Microservice frameworks on Amazon	Migration Solution	Description
Dubbo	Registry switching	Use Dubbo to switch the original service registry to EDAS Config Server and develop a new set of applications before deploying them to EDAS. Then, use SLB and configure domain names to switch traffic. Refer to this document for more details .
	Dual registration and dual subscription	Access the original service registry and the EDAS service registry when applications are migrated to enable mutual calling between migrated applications and non-migrated applications. Refer to this document for more details .

4.9 SERVERLESS

Alibaba Cloud Function Compute (FC) is a fully managed, event-driven compute service and is the best replacement for AWS Lambda.

Function Compute allows customers to migrate codes without any change from Lambda. It also integrates the container ecosystem. FC provides image deployment and supports a seamless switch from the original container deployment method.

Function Compute: Container Image Deployment

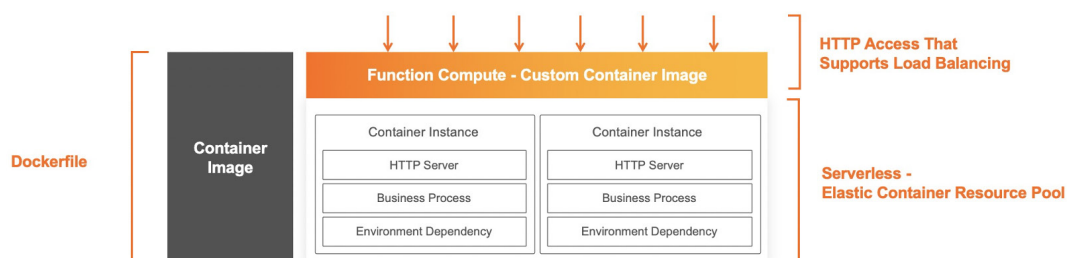
Function Compute integrates the container ecosystem. It provides image deployment and supports a seamless switch from the original container deployment method.

Disadvantages of conventional migration methods of standalone web applications (Spring Boot, WordPress, Flask, Express, and Rails):

- O&M costs are increased to cater to modernized fine-grained responsibility division and service governance.
- Historical assets such as on-cloud or off-cloud code, dependencies, and configurations make it difficult to go serverless.
- You need to plan capacity and build O&M and monitoring systems.
- Exclusive resources are allocated to low-traffic services, which causes low resource usage.

Advantages of application migration based on Function Compute and container images:

- This method migrates standalone applications at low costs.
- This method does not require O&M.
- This method supports auto scaling. You do not need to arrange capacity.
- This method ensures that all resources are in use, achieving 100% resource usage and reducing IT costs.

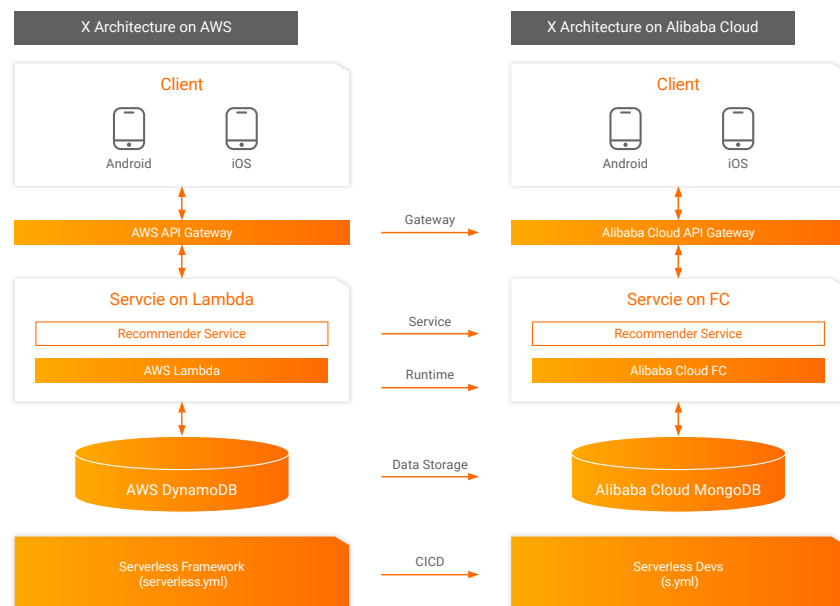


Serverless integrates the capabilities of the cloud ecosystem based on SDKs and APIs. Especially FC/Lambda, which are both integrated into the cloud ecosystem based on the unique event-driven callbacks.

As shown below, Alibaba has the most complete product matrix compared with AWS, so if the customer's services depend on other cloud products, the migration will also be easy.



	 Alibaba FC	 AWS Lambda	 Tencent SCF
Storage	• OSS • Log Service	• S3 • CloudWatch • Logs	• COS • CLS
NoSQL DB	• TableStore	• DynamoDB	- Not supported
Messaging	• Kafka • RocketMQ • MNS queue • MNS Topic	• Kafka • SQS • SNS Topic • RabbitMQ	• Kafka
IoT	• IoT Platform • IoT Link Edge • MQTT	• AWS IoT Rules Engine • Green Grass	- Not supported
Event Hub	• EventBridge	• EventBridge	- Not supported
CDN Origin	• Built-in support	• Via API Gateway	• Via API Gateway
Monitoring & Operations	• Cloud Monitor • OOS	• CloudWatch events • System Manager	- Not supported
Orchestration	• Serverless Workflow • ROS	• Step Functions • Cloudformation	• Application Service Workflow (in Beta, no international version)
Outbound Integrations	• EventBridge • FC • MNS Queue • MNS Topic	• EventBridge • Lambda • SQS • SNS	- Not supported



4.10 MESSAGE QUEUES

4.10.1 MESSAGE QUEUE FOR RABBITMQ

A distributed, fully managed, and professional messaging service that features high throughput, low latency, and high scalability. This service supports Advanced Message Queuing Protocol (AMQP) and is fully compatible with open-source RabbitMQ.

It also supports clients for different programming languages. The service is out-of-the-box and frees you from deployment and O&M. It allows you to easily migrate your business to the cloud while providing reliable and secure services.



1. Comparison between Message Queue for RabbitMQ and Amazon MQ

Category	Item	Message Queue for RabbitMQ	Amazon MQ
Performance	Peak transactions per second (TPS) of clusters	Not limited. Message Queue for RabbitMQ clusters adopts a distributed architecture that involves no primary nodes. This way, the number of nodes in use can be increased or decreased based on business needs, and thus the peak TPS of clusters is not limited.	Limited. To improve the performance of open-source RabbitMQ clusters, you must upgrade the specifications, which have upper limits.
	Peak TPS of a single queue	Not limited. Message Queue for RabbitMQ supports the increase of nodes in a single queue. No limit is imposed on the node concurrency and a queue can have an unlimited size.	Limited. The peak TPS of a single queue allowed is the peak TPS of a node allowed in the queue.
	Number of connections	Not limited. The more nodes a Message Queue for RabbitMQ instance has, the more connections the instance supports.	Limited. A single cluster supports only a limited number of connections and the threshold cannot be adjusted to a higher value. This is due to the limit on cluster performance.
	Scheduled messages	Supports message scheduling that is accurate to seconds, high-performance, and out-of-the-box.	Supports message scheduling, which can however be complex to use.
Stability	Message accumulation	Supports a large number of accumulated messages and provides high performance even in scenarios of large message accumulation.	Cannot support a large number of accumulated messages. Breakdowns may occur due to memory overuse in these scenarios.
	Scalability	Adopts a distributed architecture that involves no primary nodes. The number of nodes in use can be increased or decreased with efficiency.	Requires you to modify the cluster specifications to adjust the cluster performance.
	Service availability	Adopts a distributed architecture that ensures the high availability of 99.95% of the clusters and provides high availability across Zones.	Developed by using the Erlang programming language and requires advanced O&M experience. Availability cannot be ensured due to the open-source architecture.
	Data reliability	Uses the triplicate mechanism, which ensures high TPS performance.	Cannot ensure high TPS performance if you configure multiple replicas.
	Inspection system	Automatically detects and fixes issues, such as deadlocks and breakdowns.	Not supported.
General features	Support for client SDKs	Supports the open-source SDK for each language and each SDK version.	Supports open-source SDKs.
	Scheduled messages	Supports message scheduling that is accurate to seconds. You can use the x-delayed-message plug-in or specify the time-to-live (TTL) value to enable message scheduling.	Requires you to install a plug-in or transfer expired messages to support delayed messages.

Category	Item	Message Queue for RabbitMQ	Amazon MQ
General features	Transactional messages	Not supported.	Supported.
Exchange-related features	Exchange types	Supports direct, fanout, headers, topic, and x-delayed-message exchanges.	Supports direct, fanout, headers, topic, and x-delayed-message exchanges.
	Persistence	Supports both persistent and non-persistent storage of configurations.	Supports both persistent and non-persistent storage of configurations.
	Auto-deletion	Supported.	Supported.
	Internal	Supported.	Supported.
	Alternate exchange	Not supported.	Supported.
Queue-related features	Queue type	Adopts a distributed architecture that requires no manual configuration of the queue type and provides high availability.	Requires manual configuration. Valid types: » Classic: classic image queue. » Quorum: quorum queue.
	Node	Requires no manual configuration and no O&M operations.	Requires manual configuration. You can select a node as needed.
	Persistence	Supports both persistent and non-persistent storage.	Supports both persistent and non-persistent storage.
	Maximum length	Requires no manual configuration and supports large message accumulation.	Requires manual configuration to prevent breakdowns caused by memory overuse in scenarios of large message accumulation.
	Maximum length (in bytes)		
	Maximum memory size		
	Maximum memory size (in bytes)		
	Retry limit	Requires no manual configuration. By default, a maximum of 16 retries is allowed. For more information about retries, see Message retry .	Requires manual configuration.
	Dead letter exchange	Supported.	Supported.
	Dead letter routing key	Supported.	Supported.
	Single active consumer	Not supported.	Supported.

2. Choose instance type for Message Queue for RabbitMQ

By opening the Overview page in RabbitMQ console, you can see how to choose instance with the appropriate peak TPS.

3. Choose Migrate Amazon MQ to Message Queue for RabbitMQ



- » Migrate the metadata of the RabbitMQ cluster to the Message Queue for RabbitMQ instance, using [these detailed steps](#).
- » Migrate a RabbitMQ cluster to a Message Queue for RabbitMQ instance in dual-read mode:
 - Enable a new consumer for the Message Queue for RabbitMQ instance to consume messages from the instance.
 - Enable a new producer for the Message Queue for RabbitMQ instance, deprecate the original producer of the RabbitMQ cluster, and enable the original consumer to continue consuming messages from the cluster.
 - After all the messages in the RabbitMQ cluster are consumed, deprecate the original consumer and the RabbitMQ cluster.

4.11 DEVOPS AND AUTOMATION (TERRAFORM)

Terraform (TF) is an infrastructure as code (IaC) tool that allows you to build, change, and version infrastructure safely and efficiently. This includes low-level components such as compute instances, storage, and networking, as well as high-level components such as DNS entries, SaaS features, etc. Terraform can manage both existing service providers and custom in-house solutions.

The official Alibaba Cloud team is responsible for the daily development and maintenance of Terraform Provider. If you have used AWS/Azure/GCP (Google Cloud Platform) providers before, there will not be much change in the main framework, syntax, and usage habits. The main difference is the definition of products, resources, and business use. You can obtain detailed information about these contents through the [Alibaba Cloud official documents](#).

The following table prepares the correspondence between commonly used products and resources for you so that you can quickly compare and find the corresponding document links.

Alibaba Cloud Product Name	TF Product/TF Resource Sample	AWS Product Name	TF Product/TF Resource Sample
Container Service for Kubernetes (ACK)	Container Service for Kubernetes (ACK) /alicloud_cs_autoscaling_config	Amazon Elastic Kubernetes Service (EKS)	EKS /aws_eks_fargate_profile
Container Registry (ACR)	Container Registry (CR) /alicloud_cr_ee_instance	Amazon Elastic Container Registry (ECR)	ECR /aws_ecr_repository



Alibaba Cloud Product Name	TF Product/TF Resource Sample	AWS Product Name	TF Product/TF Resource Sample
Function Compute (FC)	Function Compute Service /alicloud_fc_alias	AWS Lambda	Lambda /aws_lambda_alias
AnalyticDB for MySQL	RDS /alicloud_db_instance	Amazon Redshift	Redshift /aws_redshift_cluster
ApsaraDB for Redis	Redis And Memcache (KVStore) /alicloud_kvstore_instance	Amazon ElastiCache for Redis	ElastiCache /aws_elasticache_cluster
ApsaraDB RDS for MySQL	RDS /alicloud_db_instance	Amazon RDS for MySQL	RDS /aws_db_instance
Elastic Compute Service (ECS)	ECS /alicloud_instance	Elastic Compute Cloud (EC2)	EC2 /aws_ami_launch_permission
Auto Scaling	Auto Scaling(ESS) /alicloud_ess_alarm	Auto Scaling	Application Autoscaling /aws_appautoscaling_policy
Resource Orchestration Service (ROS)	ROS /alicloud_ros_template	CloudFormation	CloudFormation /aws_cloudformation_stack
Virtual Private Cloud (VPC)	VPC /alicloud_network_acl_attachment	Amazon VPC	VPC /aws_default_network_acl
Server Load Balancer (SLB)	Classic Load Balancer (CLB) /alicloud_slb	Elastic Load Balancing	Elastic Load Balancing (ELB Classic) /aws_app_cookie_stickiness_policy
VPN Gateway	VPN /alicloud_vpn_customer_gateway	Site-to-Site VPN	VPC /aws_vpn_connection
Anti-DDoS	Anti-DDoS Pro /alicloud_ddosbgp_instance	AWS Shield - Advanced	Shield /aws_shield_protection
Web Application Firewall (WAF)	Web Application Firewall(WAF) /alicloud_waf_instance	AWS WAF	WAF /aws_waf_byte_match_set
Resource Access Management (RAM)	RAM /alicloud_ram_access_key	AWS Identity and Access Management (IAM)	IAM /aws_iam_access_key
Elastic Block Storage (EBS)	ECS /alicloud_disk	Amazon Elastic Block Store (EBS)	EC2 /aws_ebs_snapshot_ids
Object Storage Service (OSS)	OSS /alicloud_oss_bucket	Amazon Simple Storage Service (S3)	S3 /aws_s3_access_point
File Storage NAS	Network Attached Storage (NAS) /alicloud_nas_access_group	Amazon Elastic File System (EFS)	EFS /aws_efs_backup_policy
Log Service	Log Service (SLS) /alicloud_log_alert	AWS Cloud Watch	CloudWatch /aws_cloudwatch_composite_alarm

For more information about Terraform hands-on, please refer to:
<https://github.com/alibabacloud-howto/terraform-templates>.



REFERENCES

- 1 Gartner Market Share: IT Services, 2020

ABOUT

Established in 2009, Alibaba Cloud (alibabacloud.com), the digital technology and intelligence backbone of Alibaba Group, is among the world's top three IaaS providers, according to Gartner. It is also the largest provider of public cloud services in China, according to IDC.

Alibaba Cloud provides a comprehensive suite of cloud computing services to businesses worldwide, including merchants doing business on Alibaba Group marketplaces, start-ups, corporations and public services.

Alibaba Cloud is the official Cloud Services Partner of the International Olympic Committee.

www.alibabacloud.com/contact-sales